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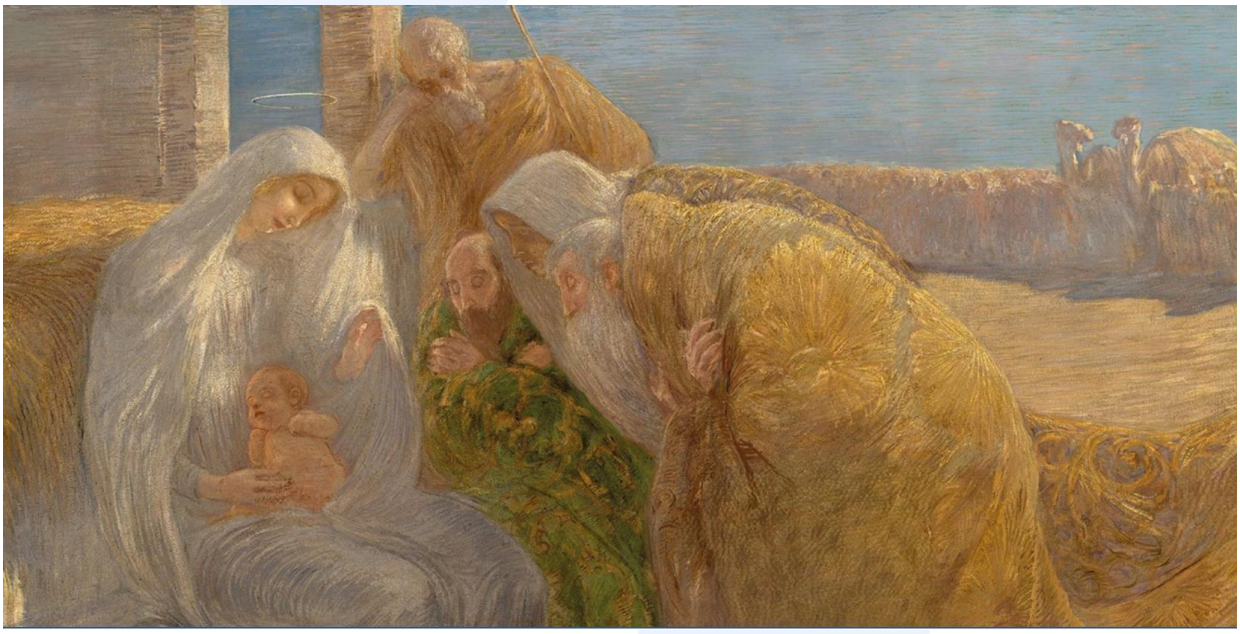
Joint Research Centre

Art{@|×}ECML

Using technology to reconcile art and science for crises - an exhibition at the European Crisis Management Laboratory (ECML)

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2025



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The Joint Research Centre: EU Science Hub
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JRC143475
EUR 40497

PDF ISBN 978-92-68-32693-0 ISSN 1831-9424 doi:10.2760/3531938 KJ-01-25-527-EN-N

Luxembourg: Publications Office of the European Union, 2025

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How to cite this report: Santini, M., Astarita, S., Borelli, G., Bortolamei, F., De Girolamo, L. et al., *Art{@|×}ECML - Using technology to reconcile art and science for crises - an exhibition at the European Crisis Management Laboratory (ECML),* Publications Office of the European Union, Luxembourg, 2025, https://data.europa.eu/doi/10.2760/3531938, JRC143475.

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Abstract

On 16 December 2024 the European Crisis Management Laboratory (ECML) launched the innovative Art{@|×}ECML exhibition, showcasing how cutting-edge technologies can help preserve and interact with cultural heritage. It went on every working day at the ECML (JRC Ispra, Italy) between 16.00-17.00 until 28 February 2025. Visitors experienced the connections between science, art and technology, by exploring potential adaptation of the technology used in the exhibition for future applications in disaster risk management science and beyond. At the heart of the exhibition was the collaboration with Haltadefinizione, the tech company of the Gruppo Panini Cultura specialised in high-resolution art digitalization. Using their Coosmo Digital Asset Management system, visitors explored world-famous art pieces on the ECML's touch video wall. This interactive display allowed visitors to zoom into fine details that are almost invisible to the naked eye, providing fresh perspectives on the iconic artworks displayed and the opportunity for a personalised navigation of the iconic art pieces.

By bringing this technology to the European Crisis Management Laboratory the exhibition bridged cultural storytelling with technological innovation. While not directed connected with the work currently done at the laboratory, the aim of the experiment was to assess the potential application of this technology to the systems and data (and also other types of art) related to disasters and emergencies.

The exhibition was enhanced by AI-generated music, with two models - ChatGPT and MusicGen – used to create a track tailored to each artwork. ChatGPT described the music that would match the tone and content of the art pieces, and MusicGen used its reply to create an instrumental composition for them.

By applying this or a similar technology to the ECML's tools, this pilot could pave the way for:

- Enhanced tools for analysing and sharing crisis-related imagery;
- Studying crisis perception through collaborations with behavioural experts;
- Preserving cultural heritage, documenting how crises impact art.

Can such an immersive experience become a starting point for a journey that takes visitors closer to a place where a crisis was or is happening? Could taking them closer make them see more? Would they change their perspective? The exhibition explored synergies between art, technology, crisis management and other scientific domains.

Acknowledgements

The successful completion of the project has highlighted the significant contributions of various individuals and organisations. Appreciation is expressed to those who have played a crucial role in its achievement. Specifically, acknowledgement is given to Haltadefinizione, Elisabetta Mastri, and Don Paolo Croci for their valuable support. The efforts of the entire ECML team, including all team members, are also recognised for their management of exhibition opening hours and dedication to their shifts. Gratitude is extended to JRC colleagues for their encouragement and assistance throughout the project. The interest and engagement of exhibition visitors, who provided thoughtful feedback, are also appreciated. Their collective support, feedback, and participation have been essential to the exhibition's success. The time and energy invested by all contributors have been instrumental in achieving this outcome. The contributions of all parties involved have been vital to the project's success.

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1 Introduction

The European Crisis Management Laboratory (ECML) of the Joint Research Centre (JRC) supports the Emergency Response Coordination Centre (ERCC) of the European Commission's Civil Protection and Humanitarian Aid Operations department (DG ECHO) and the European Commission services in general, as well as the EU agencies and the European External Action Service by providing rapid scientific analyses and seamless situational awareness on natural and human-induced disasters. With its scientific expertise, data, models, analyses and operational services, the ECML works together with the EU's operation centres to anticipate, forecast, monitor, and assess the impact of upcoming or ongoing crises, so to inform the operational decision making with scientific evidence and insights. Additionally, the ECML functions as the entry point of JRC-wide knowledge in times of crisis to provide cross-sectoral analyses.

To perform effectively the above, the ECML conducts scientific research in the field of disaster risk and crisis management. Its laboratory nature allows for experimentation in the domain of crisis management, resulting in the continuous improvement of its methodologies, modelling approaches, information systems for situational awareness, improve the methodologies for all, in accordance with the scientific state-of-the-art. The main research lines of the ECML at the time of writing are:

1. Situational awareness for complex crises,
2. Interoperability and intelligence gathering,
3. Event's modelling for anticipation, early warning and analysis of real-life events,
4. Preparedness and readiness (e.g. scenario building, exercises, community preparedness programmes),
5. Testing of innovative technology for crisis management, including AI and research outcomes.

In the area of technological studies in particular, in more than 15 years of research, the technologies explored by the ECML for their application to the field of Disaster Risk Management count Content Management Systems, drones, mobile devices, Human-Computer Interfaces, Internet of Things, virtualisation and cloud, communication and collaboration platforms, Artificial Intelligence, Augmented/Virtual Reality[1]. These technologies were explored from both a methodological and a technological point of view, for supporting the operations of a crisis room or similar facility.

This document reports the technological experiment conducted in the ECML from 13 December 2024 to 28 February 2025 and inspired by a digital technology application to art, encountered in the occasion of a visit to the art exhibition "*L'arte torna in centro*"[2] (art comes back to town) in Camerino, a medieval Italian town devastated by several earthquakes in the course of its history, the most recent ones in 1997 and 2016.

While still dealing with the recovery from the 2016 earthquake, in November 2024, Camerino municipality organised this art exhibition, successfully merging art and technology to address the emotional and cultural toll of crises. The cultural heritage of this medieval Italian town was digitalised using a Gigapixel technology by Haltadefinizione, an Italian company specialised in high-resolution art digitalisation. In his way, the local damaged treasures were brought back to life, demonstrating how technology can preserve and reimagine art in the face of disaster.

[1] Examples available here: JRC100880, European Module for Crisis Management A Crisis Room on the field Design of a ready-to-deploy mobile module for Crisis Management; JRC138914, Artificial Intelligence applied to disasters and crises management

[2] https://www.comune.camerino.mc.it/vivere-il-comune/eventi/larte-torna-in-centro/

The ECML team decided to test the interoperability of Haltadefinizione's Gigapixel technology with its hardware and software systems, while also showcasing how cutting-edge technologies can preserve and interact with cultural heritage. Using the Haltadefinizione Coosmo Digital Asset Management system, or Coosmo, the Digital Asset Management system specially designed to manage digital artistic and historical heritage in ultra-high definition, visitors could explore world-famous art pieces on the ECML's cutting-edge touch video wall. This interactive display allowed users to zoom into fine details almost invisible to the naked eye, providing a fresh perspective on the iconic artworks displayed.

By bringing this technology to the European Crisis Management Laboratory, the exhibition bridged cultural storytelling with technological innovation. The stories in the exhibition had very little to do with the crisis management work done at the laboratory. Instead, this was an experiment to assess the potential of applying the same technology to the systems and data (and also other types of art) that relate to disasters and emergencies.

2 Overview of the Art{@|×}ECML exploratory project

The name "Art{@|×}ECML" for the exhibition was chosen to suggest the double nature of this experiment. The brackets in the exhibition title indicate that either symbol can be used in the reading, so that the title had two meanings: Art@ECML, i.e. art at the ECML, and Art×ECML, i.e. art for the ECML, highlighting the double purpose of the project: bringing art in through technology and using art technologies at the ECML.

The exhibition lasted from 13 December 2024 to 28 February 2025, being open every working day from 16:00 to 17:00 for visitors from inside the Joint Research Centre as well as external visitors (in this latter case, upon invitation). Staff working at the ECML team introduced the experience to visitors, supporting their technological needs and replying to questions.

The participants experienced the connections between science, art and technology in an experiment to explore how to adapt the technology used in the exhibition to future applications in disaster risk management and other JRC research fields.

The exhibition was enhanced by AI-generated music. We combined two AI models, ChatGPT by OpenAI and MusicGen by Meta, to produce a track tailored to each artwork. ChatGPT described the music that would match the tone and content of the art pieces, and MusicGen used the generated description as input to create an instrumental composition. MusicGen was trained only on licensed music and its outputs are here used in accordance with its Creative Commons Attribution-NonCommercial 4.0 International Public License[3].

Each visitor was then provided with an immersive experience and was also asked to share their ideas about possible applications of the Gigapixel technology to crisis management or to their own research fields, from the wide range of JRC research.

The exhibition featured the following art pieces, available for free navigation in high resolution on the Haltadefinizione website. Details of the above-mentioned art pieces are provided in Annex 1.

- Adoration of the Magi (Adorazione dei Magi) c.1423 Gentile da Fabriano[4]
- Adoration of the Magi (Adorazione dei Magi) c.1896 Gaetano Previati[5]
- Spring (Primavera) c. 1482 by Sandro Botticelli[6]
- The Marriage of the Virgin (Lo Sposalizio della Vergine) c.1504 Raffaello Sanzio[7]
- Scrovegni Chapel Frescoes (Affreschi della Cappella degli Scrovegni) 1303-1305 Giotto di Bondone[8]

Ideally, incorporating art pieces that depict natural hazard events—such as the eruption of Mount Vesuvius (a frequent subject in artworks), earthquake damage (as illustrated in Giotto's frescoes in the S. Francesco chapel in Assisi), or tsunami scenes rendered by artists worldwide—would have strengthened the project's alignment with the ECML's focus areas. However, the selection of artworks was limited by two factors: the availability of relevant pieces in the Haltadefinizione archives and the

[3] License: https://huggingface.co/spaces/facebook/MusicGen/- blob/main/LICENSE_weight

[4] https://www.haltadefinizione.com/visualizzatore/opera/adorazione-dei-magi-gentile-di-niccolo-di-giovanni-gentile-da-fabriano#scroll

[5] https://www.haltadefinizione.com/visualizzatore/opera/adorazione-dei-magi-gaetano-previati

[6] https://www.haltadefinizione.com/visualizzatore/opera/primavera-sandro-botticelli

[7] https://www.haltadefinizione.com/visualizzatore/opera/sposalizio-della-vergine-raffaello-sanzio-1

[8] https://www.haltadefinizione.com/en/image-bank/giotto-di-bondone-scrovegni-chapel-360-view/

tight timeline for preparing the exhibition. Despite these constraints, each chosen artwork was successfully linked to a scientific field, as outlined in Annex 2.

3 Project objectives

The vision of the "Art{@|×}ECML" project was to transform temporarily the ECML into a space where science, technology and art can inform and elevate each other. This exhibition conceived out of a shared ambition to explore the synergies between art, technology and crisis management. The ECML team saw an opportunity to adapt Haltadefinizione's technology to their own tools and products, such as high-resolution satellite imagery and other disaster management data. Despite the artworks in the exhibition not being related to the work done at the ECML, this initial trial might pave the way for improvements in how crisis-related images are displayed, analysed, and shared, studies about the ways citizens perceive and depict disasters, and contributions to ongoing projects about the impact of crises on art (e.g. Syrian conflict's effect on cultural artefacts).

For the ECML, the aim of the exhibition was to explore synergies between art, technology and crisis management. By applying this or a similar technology to the ECML's tools, it was envisaged that this pilot could have paved the way for:

- Enhanced tools for analysing and sharing crisis-related imagery;
- Studying crisis perception through collaborations with behavioural experts;
- Preserving cultural heritage and documenting how crises impact art.

From a broader perspective, the aim was to conduct a two-months-long brainstorming at the JRC to collect ideas on how this digital technology could serve the scopes of other research fields. By exploring the connections between science, art, and technology, JRC scientists who visited the exhibition could reflect on and share their ideas on how to use this technology in their domain of expertise.

Beyond the main objective of testing of the Gigapixel digital technology in the ECML, additional objectives included:

- Increasing the use of the video wall (a similar one is being installed in the ERCC[9]);
- Inspiring different and innovative ways to communicate science;
- Attracting colleagues to the ECML, thus increasing its visibility;
- Experiencing the collaboration with a digital tech company, testing how it works if the ECML was to become a JRC Open Lab;[10]
- Enjoying an immersive experience of masterpieces from the past, thus contributing to the personnel's well being.

[9] Emergency Response Coordination Centre of the European Commission's Civil Protection and Humanitarian Aid Operations department (DG ECHO)

[10] https://joint-research-centre.ec.europa.eu/tools-and-laboratories/open-access-jrc-research-infrastructures_en

4 Gigapixel imaging technology

Definition: A *gigapixel image* is an ultra-high-resolution photograph containing at least one billion pixels (1000 megapixels).

Such enormous resolution far exceeds standard digital photos, enabling prints many feet wide with fine detail. For example, a gigapixel image can be enlarged to mural size (e.g. 22 ft by 4 ft or 6.7 by 1.2 m) while preserving clarity down to small features, allowing viewers to zoom in and discern details (like distant faces or brushstrokes) that would be impossible to see in ordinary images (sandiegoreader.com).

Mosaic Stitching (Panoramic Approach): The most common method for creating gigapixel images is to capture many high-resolution photos of sections of a scene and stitch them into a single large mosaic. A robotic panoramic mount can automate this process by taking photographs *section by section* in a grid (rows and columns), which are later blended seamlessly into one wide-field image (bhphotovideo.com, phys.org). Each individual shot covers a small part of the scene with a telephoto or microscope lens, and overlapping shots ensure that the stitching software can align them accurately. The resulting composite image contains billions of pixels and preserves extreme detail across a wide field of view. This mosaic approach is frequently used for panoramic cityscapes, stadium crowds, and even museum exhibitions. For instance, the Mauritshuis museum in the Netherlands created an interactive *"gigapixel museum"* by capturing *countless small images* of each gallery with 360° cameras and stitching them together into an immersive high-resolution tour (artstalkmagazine.nl). Likewise, Google's Art Camera uses a robotic system to capture *hundreds of close-up photos* of a painting and then merge them into a single gigapixel image of the artwork (shutterbug.com). Finally, this is also the technology Haltadefinzione used for our own exhibition. The mosaic technique has become the backbone of gigapixel photography due to its relative accessibility – even consumer Digital Single-Lens Reflex (DSLRs) can contribute to multi-billion-pixel panoramas when used with motorized pano-heads and stitching software.

Alternative Capture Techniques: Besides mosaicking, several other methods can achieve gigapixel-scale images:

- **Large-Format Scanning Cameras:** One approach is to use an ultra-large analog film or sensor and then digitise or scan it at extremely high resolution. The Gigapxl Project (led by Graham Flint in the early 2000s) exemplified this by using custom-built large-format film cameras. Flint's camera exposed 9″×18″ sheet film with a specially designed 8-element *"Asymmagon"* lens, then scanned the film at high DPI to produce digital images on the order of 4 gigapixels (sandiegoreader.com). This method relies on a single enormous exposure rather than many small ones, but it demanded significant engineering in lens design and film scanning to push beyond the gigapixel threshold.

- **Multi-Sensor Array Cameras:** Another approach uses an array of many sensors and lenses capturing sub-images simultaneously, which are then combined computationally. An example is Duke University's **AWARE-2** camera, a gigapixel digital camera composed of 98 micro-cameras behind a shared spherical objective lens (petapixel.com). Each micro-camera (with a 14-megapixel sensor) photographs a portion of the field, and a central processor stitches these into one contiguous image. This design effectively functions as a parallel multi-camera system, bypassing the need for sequentially taken mosaics. Multi-sensor array cameras can capture a gigapixel snapshot in a single shutter actuation, albeit with complex hardware (the AWARE-2 prototype was ~3% optics and 97% electronics by volume to handle data and processing). Variants of this idea appear in advanced surveillance and scientific imaging systems. For instance, the **Pan-STARRS** observatory uses a

mosaic of CCD sensors in its telescope camera (1.4 gigapixels total) to capture wide sky images in one go, and the upcoming Vera Rubin Observatory's camera contains a *3.2-gigapixel* focal plane array of 189 CCDs for single-shot ultra-resolution astronomy (lsst.org). These multi-sensor systems achieve gigapixel images without post-capture stitching, though they require careful alignment and calibration of many optical channels.

- **Scanning and Line-Sensor Cameras:** High-resolution scanning back cameras and rotating line-sensor panoramic cameras can also produce gigapixel imagery. In these systems, a single linear or medium-format sensor is moved across the image plane or scene to build up a large image strip by strip. For example, some panoramic photographers use motorised rotation coupled with a scanning sensor to capture 360° images with extremely high pixel counts. This approach is conceptually similar to mosaicking, but the sensor motion (rather than discrete frames) generates the continuous high-res image. Such systems were more common in specialized applications (aerial or orbital imaging) and have largely been supplanted by the mosaic method and multi-sensor designs for ground photography.

In practice, the mosaic approach remains the most popular for gigapixel photography in fields like tourism, sports, cultural heritage and virtual tours, due to the availability of robotic mounts and software. Exotic one-shot solutions (multi-sensor or custom cameras) are mostly found in research, military, or scientific contexts where real-time gigapixel capture is needed. Each method has trade-offs: mosaics are accessible but require time and stitching computation, whereas custom large-format or array cameras capture instantly but involve complex hardware and expense. Further details on the Gigapixel imaging technology are provided in Annex 4 to 7.

5 Technical details of the Haltadefinizione Coosmo Digital Asset Management system

Panini Cultura is a network of companies[11] offering a range of complementary and synergistic services for the protection, conservation, use and enhancement of cultural heritage. Thanks to this ecosystem, Panini Cultura covers the entire Digital Humanities supply chain: from the digitisation of archives and cultural objects, to metadata, logistics, conservation and file management, to their enhancement through editorial products, interactive digital systems and online promotion.

Coosmo is the Digital Asset Management service of Haltadefinizione, designed to simplify the management and immediate use of gigapixel images and more. The platform provides digital archiving and digital preservation services for cultural heritage, ensuring the protection and security of assets for both public and private institutions and entities. Specifically developed to handle large volumes of digital content, Coosmo is fully compatible with IIIF Image and Presentation API standards and can be easily adapted to multiple needs, from preservation to enhancement. With advanced online and offline tracking systems, it guarantees that images are always protected, accessible, and quickly available. Its intelligent storage management allows even extremely large files, such as gigapixel images, to be delivered via cloud through IIIF and DZI protocols, with the same quality and performance as standard-sized images.

They produce digital reproductions for public archives and corporate archives, libraries and photographic collections. Through digitisation, they guarantee long-term preservation and easy access to documents, while ensuring the protection of the originals. The quality of their reproductions allows for the enhancement of cultural and corporate heritage and assets, the printing of facsimiles and the creation of a single file for consultation and metadata. The reproduced images allow for the consultation of documents that would otherwise be inaccessible, a longer life for the original and its reuse for scientific and commercial purposes.

Of the portfolio or their products -which includes documents digitisation, 3D digitisation of cultural and museum objects, digitisation of environments and monuments, Virtual Reality (VR) and Mixed Reality (MR) experiences starting from the processing of our gigapixel images and 3D models- in this project we explored:

- The output of Scanners for cultural heritage and their laboratories for the digitisation of cultural heritage
- Coosmo Digital Asset Management
- The Touch experience & digital stories product.

The company designs, engineers and industrialises scanners and photographic technologies for 2D and 3D acquisition. The experience they have gained in digitising documents and works of art has led them to develop custom hardware solutions for digitising diverse heritage assets, with cutting-edge technical specifications that exceed the capabilities of standard tools available on the market.

C56 is the headquarters of the Panini Cultura Group, an interdisciplinary centre for innovation and research where the Group's companies collaborate on the enhancement and digital transformation of

[11] Haltadefinizione (Extraordinary digitisation, replica valorisation platform and digital twin); Hyperborea (software and services for archives); Memooria (hardware and software for historical heritage digitisation); (Franco Cosimo Panini (e-commerce merchandising, specialised publishing, exhibitions and events, museums' bookshops); Midadigit (big digitalisation projects).

cultural heritage. The hardware for digitising works of art in the laboratory is developed by Memooria, a company specialising in the design and engineering of innovative technologies for digital transformation in the cultural heritage sector. The footage and images processed by these advanced machines can take the eye inside the subject, beyond physical boundaries.

Thanks to their proprietary technologies, they create gigapixel images, high-resolution digital reproductions obtained by combining multiple images of details of the same subject. The process is presently arriving at a density of 5000 ppi (points per linear inch), where a pixel is approximately 0.005 mm in both dimensions. Since at least two pixels are needed to discriminate and see a feature, the details can be as little as 0.01 mm, that allows scanning microfilms. The quality standard adopted is ISO 19264-1. The result of this process guarantees extraordinary definition capable of showing even the smallest details, offering new opportunities for dissemination, enjoyment, study and research.

Tactile experiences and digital storytelling are ideal for enriching cultural narratives and improving interactivity. Using touchscreens, interactive surfaces and apps, users can explore content in a personalised and in-depth way.

6 Technical details of the ECML video wall and software

The ECML Crisis Room has a large visualisation device with 30 Mpixels resolutions spreading across 12.4 m^2. Technically, it consists of 15 BARCO Unisee 700, modular LCD displays which help reduce to the minimum the gaps between the panels and to maximise the alignment of the panels in terms of luminosity and colours. SenseX, a sensor technology which provides brightness and colour stability over time and across the entire display wall, makes this possible.

A 32-point multitouch IR overlay is installed over the wall frame using magnetic holders. The system is connected to the Graphical controller with an USB 3.0 cable, making it the largest touch device in Europe.

The whole display area is managed by a Graphical controller, a BARCO TransForm NSD-420, running a Windows Operating System (OS). Its main features include an Intel® Xeon processor, 128 GB Ram, an SSD raid 1 disk, and a redundant power supply.

It virtualises the whole display in a single video output, as long as the OS is concerned; therefore, it is easy to manage all contents a PC can display.

The controller is provided with an installation of BARCO CMS, a content management system used to present and organise contents on the display. The specificity of the Disaster Risk Management domain requires a software more specific than the general-purpose one provided with the devices and more dynamic than the typical plant surveillance, as discussed in Collaborative Human-Computer Interaction with Big Wall Displays - BigWallHCI 2013 3rd JRC ECML Crisis Management Technology Workshop[12].

JRC MuSCLE (Multi Source Crisis Layout Engine) is a client/server application allowing operators to arrange swiftly on the wall the contents they are visualising on their displays.

Starting from the general approach of Barco CMS (Content Management System), a collection of rectangular objects composing the visualisation and hosting different information sources, the JRC MuSCLE offers an agile interface and a wider set of sources that allows a richer experience. With the use of Microsoft Edge WebView2 component, it can display fully browsable web pages.

This functionality is used to present the contents displayed by the Haltadefinizione web components, that are javascript based and can offer several user experiences such as gigapixel display, 360° display with hot spots, story-based hot-spot visualisation, side-by-side and overlapping visualisation.

The Haltadefinizione embedding link is used together with an authorisation token to access directly the chosen image and display it full screen.

JRC MuSCLE is specifically aimed at visualising contents and has no sound-oriented components. To improve the user experience, the AI generated music was converted to a blank video and then used as soundtrack. The software positioned both on the display and played the video in loop.

To allow users navigating the inventory of selected artworks, a custom user interface was developed, providing a touch interface to choose which artwork to display on the video wall. It also allows muting the music playback and restoring the display to the exhibition presenting poster.

Furthermore, the user interface played a key role in managing the Crisis Room's displays, serving multiple purposes. It presented introductory texts for each artwork, informed passersby that the room

[12] http://publications.jrc.ec.europa.eu/repository/handle/JRC85533

was open for the art exhibition, and seamlessly reverted all displays back to their normal operational mode once the exhibition concluded. A web-based version was developed to demonstrate the exhibition on a single PC.

Figure 1. User interface



Source: *JRC E1, Art@ECML*

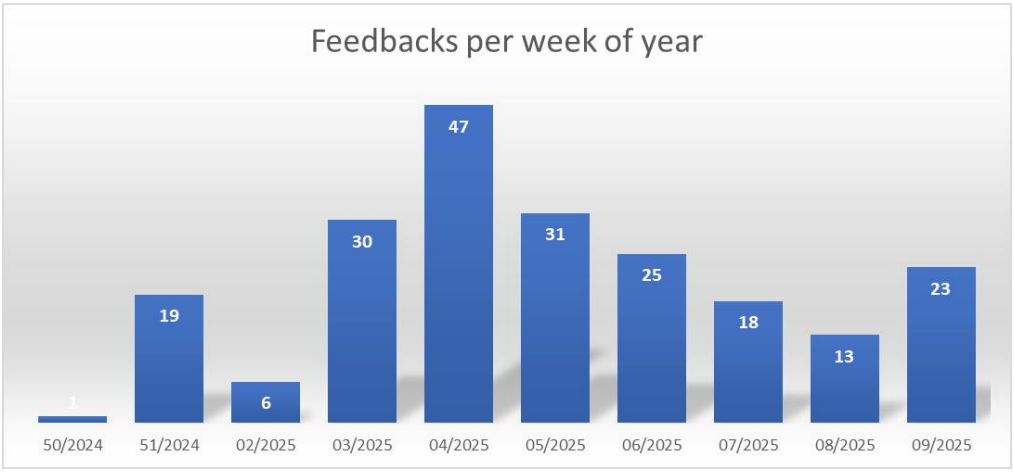
7 Outcomes of the internal brainstorming on the exploratory project

A large number of colleagues and their invited guests from outside JRC - including delegations from other DGs of the European Commission and private guests – visited the Art{@|×}ECML exhibition. Unfortunately, not all guests remembered to sign the digital visitors' book, or fill-in the dedicated EU survey. This leads to an estimated total number of visitors of approximately 600 people (despite the number of registered visitors being around 200).

7.1 Number of visitors

- Total registered visitors: 212
- Visitors who left feedback: 149

Figure 2. Number of visitors per week of the year (week/year)

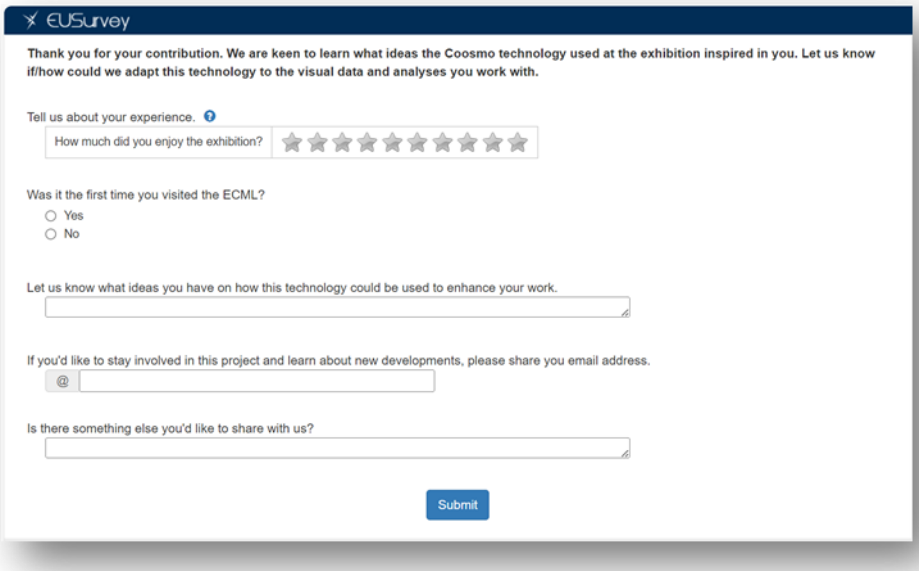


Source: *JRC E1, Art@ECML*

An EU Survey[13] was designed to collect visitors' feedback, together with a 3D geo-located guest book of the exhibition, where they could pin their location and leave comments and ideas.

[13] https://ec.europa.eu/eusurvey/runner/d373cae3-f0b3-9cba-36a0-44402fc6f840

Figure 3. Feedback survey



Source: *JRC E1, Art@ECML*

Through the survey we collected 21 feedbacks with an average evaluation of 9.3 out of 10 and almost every participant left ideas or comments.

7.2 Survey outcomes

The survey outcomes highlight various potential innovative uses for ultra-high resolution imaging technology. Visitors mentioned they would be eager to explore applications in areas such as urban planning, warfare strategy, and crisis management by visualising natural patterns. In healthcare, there was excitement about using the technology to enhance digital pathology and research, especially in understanding diseases like cancer, by integrating high-resolution and spatial omics[14] data.

Visualising complex networks, such as the Digital ecosystem analysis (DGTES)[15] digital sphere, and enabling biometric identification from a distance also emerged as potential uses. Additionally, there was enthusiasm for engaging artists to experiment with the technology, creating educational experiences that blend art and science.

The technology's potential in monitoring infrastructure through high-resolution imagery for roads, bridges, and traffic flows was noted, offering improved maintenance and security management. It could also play a role in public engagement, particularly in education for disaster risk reduction by using vivid storytelling.

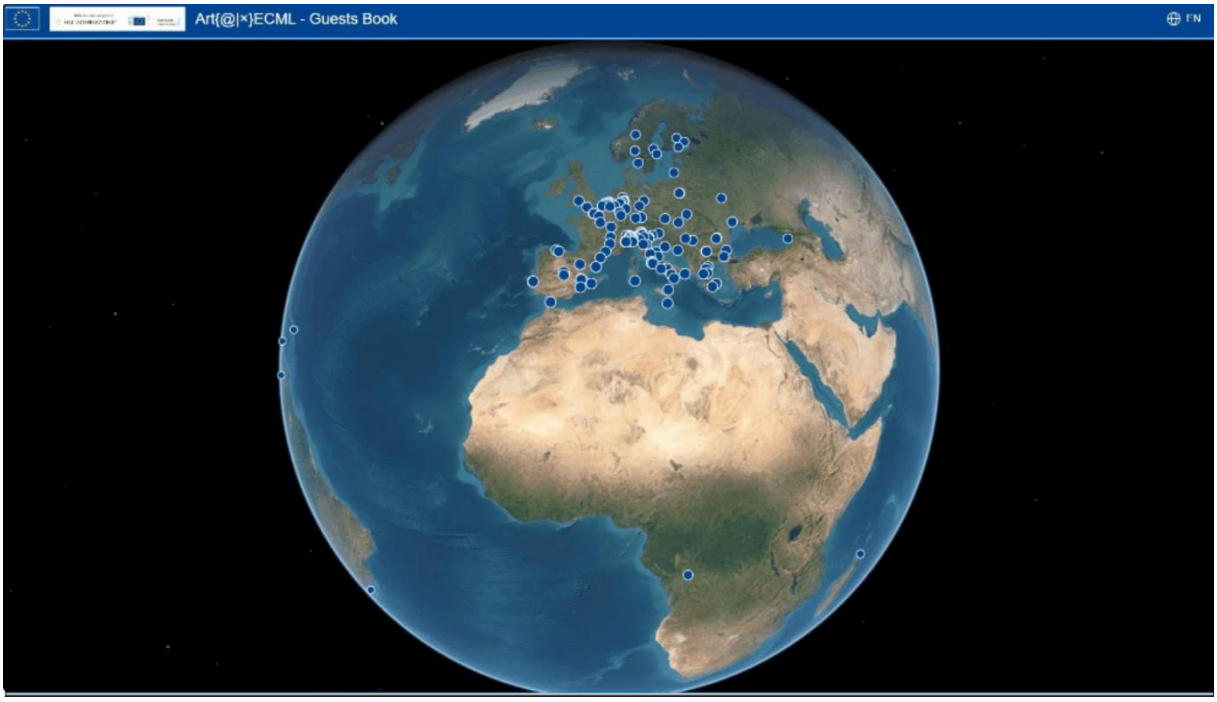
However, concerns about the technology's significant data storage needs and environmental impact were raised. Some participants feel current technologies adequately meet practical demands, questioning the necessity of deploying such high-resolution solutions.

[14] https://en.wikipedia.org/wiki/Omics

[15] https://joint-research-centre.ec.europa.eu/predict/digital-ecosystem-analysis-dgtes-2022_en

Overall, while impressed by the technology's potential across various fields, respondents emphasise the need to address environmental and ethical considerations, highlighting a balance between innovation and existing solutions.

Figure 4. Geolocated virtual guest book



Source: *JRC E1, Art@ECML*

7.3 Guests book outcomes

During the exhibition about 190 visitors left a message or a pin in the ART{@|x}ECML guests' book. 131 visitors left feedback about the exhibition, sharing impressions, suggestions and, in some cases, even feelings.

Overall, the feedback was highly positive. Users expressed great satisfaction with the ability to explore famous paintings in unparalleled detail, describing their experience as "amazing," "impressive," "breathtaking," and "emotionally engaging." The exhibition was praised for its combination of art and technology, and many users expressed interest in seeing the technology applied to other fields.

In the feedback, users expressed enthusiasm for various potential applications of the technology. They suggested that it could be employed in education and research, aiding the analysis of art pieces and facilitating detailed examinations in medical research and engineering inspections. Additionally, they proposed combining the technology with other datasets, such as electron microscopy images or disaster maps, to enhance the visualisation and comprehension of complex information across various disciplines. Lastly, users envisioned the integration of this technology with cultural heritage preservation efforts, as it could provide detailed and accessible documentation of historical artifacts and sites, contributing to better conservation and appreciation of our shared cultural legacy.

Some users also suggested enhancements for future exhibitions, including the expansion of the collection of high-resolution images to offer users an even more extensive exploration of renowned art pieces. They also emphasized the importance of providing information on the artists, techniques, and historical context of the displayed artworks, as this would significantly enhance the educational aspect of the experience. Lastly, users expressed interest in integrating the technology with virtual or

augmented reality, aiming to create an even more immersive experience for visitors, allowing them to engage better with the art and the technology itself.

The details of the visitor's comments and suggestions are shared in Annex 3. A summary of the research domains for potential future applications of the Giga Pixel technology is listed below:

1. Technology & Digital Systems

- Digital Techno-Economic Ecosystem
- Biometric identification
- Dynamic images (eg flows in the ocean or atmosphere)
- Pictures taken in infrared
- Spatial (geographical) data

2. Science & Research

- Digital pathology and cellular research
- Documenting artisanal practices and the products of immaterial patrimony
- Fractal nature of natural systems
- Details and anomalies in the surface of objects (e.g., reactor tanks)

3. Infrastructure & Security

- Display of Global Human Settlement Layer data
- Urban warfare planning and mitigation
- Monitoring infrastructure conditions
- Border security
- Monitoring traffic flow and congestion

4. Environment & Land Management

- Land management and carbon dynamics

5. Public Engagement & Awareness

- Public engagement and awareness
- A browsable picture for each JRC site at the Europe Science Experience (ESE) visitor centre

6. Cultural heritage preservation

- Making records that document a work's conservation status
- Finding works stolen during wars, uprisings
- Protecting national and international historical artistic heritage.

8 Desk review of examples of scientific applications of Gigapixel technology

Gigapixel imaging has emerged as a transformative tool across diverse scientific and practical domains. By capturing images with resolutions exceeding one billion pixels, gigapixel cameras enable unprecedented detail and coverage, offering solutions for challenges ranging from geohazard monitoring to wildlife conservation. Below we report some examples of multifaceted applications of gigapixel technology, highlighting its impact on science, public safety, and industry.

Geohazards Assessment and Environmental Monitoring

The GIGA2 project[16] by Italian researchers Saverio Romeo and Alessandro Fraccica exemplifies gigapixel technology's role in geohazard assessment. Using high-resolution imaging, the system tracks subtle ground movements, such as landslides or tectonic shifts, over time. By analysing gigapixel images, geologists can identify early warning signs of environmental risks, enhancing disaster preparedness and community resilience. This application is critical for regions prone to earthquakes or coastal erosion, where timely interventions save lives and infrastructure[17].

Industrial and Emergency Management

In industrial settings, gigapixel cameras provide ultra-high-definition (UHD) monitoring of vast areas, such as industrial parks or manufacturing sites. Systems like those developed by Zohetec[18] offer panoramic views and zoom capabilities, enabling real-time surveillance and detailed post-incident analysis. For emergency management, these cameras support rapid response to accidents or spills, ensuring safety for workers and the environment. Their ability to cover expansive zones with a single device reduces blind spots and improves operational efficiency.

Citywide Surveillance and Public Safety

Urban centers are adopting gigapixel cameras to enhance public safety. Zaitechno[19] highlights how these systems monitor crowded areas, traffic, and critical infrastructure, deterring crime and aiding law enforcement. By integrating gigapixel imaging with existing surveillance networks, cities can achieve comprehensive coverage, identifying suspicious activities and optimizing emergency response routes. However, ethical considerations around privacy must be balanced with the benefits of such pervasive monitoring.

Wildlife Conservation and Biodiversity Protection

Gigapixel cameras are revolutionizing wildlife research, as demonstrated by Zaitechno's projects[20]. These devices capture minute details of animal behaviour, monitor endangered species, and detect poaching activities in remote habitats. For instance, high-resolution footage can identify individual animals or track migration patterns, providing data crucial for conservation strategies. The

[16] https://misc.cdri.world/fellowship/cohort/2024-25/project/giga2-gigapixel-imaging-for-geohazards-assessment-and-awareness

[17] https://www.researchgate.net/publication/236980567_Gigapixel_Imaging_and_Photogrammetry_Development_of_a_New_Long_Range_Remote_Imaging_Technique

[18] https://www.zohetec.com/case/82-en.html

[19] https://zaitechno.com/citywide-vigilance-why-gigapixel-cameras-are-essential-for-public-safety-now/

[20] https://www.youtube.com/watch?v=tVG04ZZsNR8

technology's non-invasive nature ensures minimal disruption to ecosystems while offering insights into biodiversity dynamics.

Securing Critical Infrastructure

Critical infrastructure, such as power plants, data centers, and border facilities, benefits from gigapixel cameras' ability to detect intrusions or structural vulnerabilities. Such systems enhance security by identifying threats in real time, from unauthorized access to environmental hazards like wildfires.

Science Communication and Education

Museums and educational institutions are leveraging gigapixel imaging to engage audiences[21]. Projects like Google's Art Camera allow virtual exploration of artworks at the brushstroke level, democratising access to cultural heritage. Similarly, gigapixel microscopes, such as Duke University's 3D model, enable biologists to study cellular processes in unprecedented detail[22]. These tools bridge the gap between science and the public, fostering curiosity and understanding.

Advancing Scientific Research

Beyond practical applications, gigapixel technology drives cutting-edge research. The SLAC National Accelerator Laboratory's 3,200-megapixel camera, part of the Rubin Observatory, captures cosmic phenomena to unravel mysteries like dark matter[23]. In ecology, unbiased population size estimation using gigapixel images ensures accurate biodiversity assessments[24]. Meanwhile, VISTA's gigapixel mosaic of the Milky Way showcases the scale of astronomical imaging, offering insights into galactic structures[25].

AI and Future Innovations

Artificial intelligence is amplifying gigapixel imaging's potential. Topaz Labs' Gigapixel 8 uses deep learning to upscale images, enhancing data resolution for fields like remote sensing and medical imaging. Meanwhile, advancements in photogrammetry and long-range remote imaging expand applications in archaeology and environmental monitoring. Drones and satellites equipped with gigapixel cameras are set to revolutionize persistent surveillance and planetary science.

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[21] https://mw2013.museumsandtheweb.com/paper/exploring-gigapixel-image-environments-for-science-communication-and-learning-in-museums/index.html

[22] https://gigazoom.rc.duke.edu/team/Eagle%20(MCAS-48)/Digital%20Cytopathology

[23] https://www6.slac.stanford.edu/news/2020-09-08-sensors-worlds-largest-digital-camera-snap-first-3200-megapixel-images-slac

[24] https://pratt.duke.edu/news/3d-gigapixel-microscope/

[25] https://www.eso.org/public/images/eso1242a/

9 Discussion of the results

Art{@|×}ECML has been conceived as a temporary art and technology exhibition in the European Crisis Management Laboratory with a number of objectives:

- test the Giga Pixel digital technology in the ECML
- increase the usability of the video wall
- attract colleagues to the ECML, thus increasing its visibility
- inspire how to communicate differently our science and knowledge
- experience the collaboration with a digital tech company, thus experiencing how it could be if we work to become a JRC Open Lab
- enjoy an immersive experience of masterpieces from the past, thus contributing to the personnel well being

Primarily the idea was to test the Giga Pixel digital technology in the ECML, experimenting the interoperability of this technology with the hardware and software of the Laboratory. Displaying the Gigapixel resolution images of the art pieces selected proved to be easily achievable, thanks to the JRC-designed MUSCLE software, the BARCO videowall of the Laboratory and the Haltdefinizione Coosmo Digital Asset Management. The navigation of the art pieces by visitors via the use of the ECML touch videowall has been seamless and allowed for a personalised experience of the exhibition, with a very engaging immersive effect, as testified by the feedback messages left by the visitors.

The experience allowed the JRC to become aware of the work of Panini Cultura, i.e. a network of companies offering a range of complementary and synergistic services which covers the entire Digital Humanities supply chain: from the digitisation of archives and cultural objects to metadata, logistics, conservation and file management, to their enhancement through editorial products, interactive digital systems and online promotion. This network might be a useful one to explore other digital technologies within the ECML. For example, 3D digitisation, high resolution digital twins of buildings via aerial photogrammetry, their virtual indoor and outdoor exploration using virtual reality and augmented reality visors, the digital storytelling are all options which could be considered for exploratory projects in the domain of science communication, enriching the narratives with interactivity and immersive approaches. Some research in the direction of innovative educational experiences that integrates the person's interaction with scientific contents through an interdisciplinary approach, and the use of advanced digital tools could inspire new ways of communicating JRC science.

The project allowed to attract a good number of colleagues to the ECML, thus increasing its visibility within and beyond JRC, thanks to the visiting external delegations. Many JRC colleagues entered the laboratory for the first time, thus becoming aware of its core business, i.e. disaster risk and crisis management. Some others demonstrated enthusiasm for the initiative and left feedback with suggestions for the exploration of use of the Gigapixel technology with scientific data. The ideas collected from the visitors of the Art{@|×}ECML exhibition somehow converge with the desk review of scientific applications performed, which allowed to identify examples of applications of this technology in a vast range of domains, some of which being: geohazards assessment and environmental monitoring; industrial and emergency management; citywide surveillance and public safety; wildlife conservation and biodiversity protection; securing critical infrastructures; science communication and education in general.

All the visitors highlighted also the engaging effect of such way of exploring digital images and mentioned their appreciation for an "out of the box" initiative, which provided them with the opportunity of an immersive experience of masterpieces from the past, thus indeed contributing to a sense of wellbeing at the end of the visit. The combination of the use of the big touch-screen video

wall of the ECML with the Gigapixel images and the background AI-generated music created in fact a relaxed, contemplative atmosphere at the end of the working day (as the exhibition was open for visits between 16:30 and 17:30). The evocative power of the Art{@|×}ECML set up proved to be high, therefore this effect on individual's perception could be applied also to different domains, to effectively convey other messages (e.g. to raise awareness and suggest empathy about environmental dynamics, societal issues and other policy-related contents).

10 Project communication and outreach

The project was communicated via a few internal channels and internally distributed communication products. As for the latter, an exhibition flyer, a one-pager on the event introduction, 5 art labels (one for each of the selected art works) and 10 postcards (on specific details of the art works somehow related to a scientific domain) were designed, printed and distributed (see Annex 2). Finally, a power point presentation was created, to introduce the concept and the scope of the exhibition to visitors (see Annex 8).

Information on the event was shared only through internal communication channels within JRC, as the exhibition was accessible only by JRC personnel and invited guests from outside. news article was shared on SharePoint: https://eceuropaeu.sharepoint.com/sites/ORG-jrc-portfolio-25/SitePages/Art%7B@-%C3%97%7DECML--Using-technology-to-reconcile-art-and-science-for-crises---an-exhibition-at-the-JRC-Ispra.aspx and some posts on ViVaEngage. Finally, the internal JRC Newsletter of December 2024 featured the information on the event as well (https://ec.europa.eu/newsroom/jrcstaffcommunication/newsletter-archives/58400).

The exhibition was visited by individual colleagues but also by delegations from other Units of JRC, from JRC portfolios[26] or other Directorates of the European Commission (e.g. delegation from DGIT). To allow visitors to benefit from a professional description of the art pieces exposed - and not only to experience the effect of the Gigapixel technology and inspire their ideas on how to apply it to scientific data - the European Crisis Management Laboratory organised two seminars, with art historians:

- The seminar "Art and interdisciplinarity" with Don Paolo Croci, who holds a Degree in Modern Literature and a Bachelor in Theology. His teaching experience includes tenure as an editor at the publishing house Queriniana (Brescia, 1990-1996), serving as a literary director at the Ancora publishing house in Milan. Meanwhile, he collaborated as a subject expert in History of Liturgy and Liturgical Theology at the Catholic University of the Sacred Heart in Milan from 1987 to 1996.

The seminar focused on this question: "How many authors painted the Spring? A painter, a musician, a poet, a student of nature". The many dimensions of the artists are present in their works. Dr. Paolo Croci gave an interpretation of the Spring and the Scrovegni chapel that reveals the proficiency of their authors in many different arts and crafts.

- Seminar "Zooming In on Imperial Greed for Beauty: A Journey from Imperial Gothic to Italian Renaissance" with Elisabetta Maistri, Art Historian and Head of Collection and Educational Programmes at Eccel Kreuzer Museum in Bolzano (20 February 2025).

Between the late 18th and early 19th centuries, as Napoleon Bonaparte swept across Europe, he sought to establish a European archive of knowledge in Paris through the confiscation of artworks and archival documents. Italy was one of the unquestionable victims of the First French Empire's greed, as the vicissitudes of Gentile da Fabriano's *The Adoration of the Magi* and Raphael's *The Marriage of the Virgin* prove. However, the French war pillages are merely the pretext for our discussion, which offers a historical journey from the International Gothic to the Renaissance by zooming in on these two masterpieces, respectively from the Uffizi Gallery and the Brera Picture Gallery. This talk was made possible thanks to the increasingly popular processes of digitisation, which favour access to distant cultural objects, promote an in-depth study of the humanities remotely,

[26] https://joint-research-centre.ec.europa.eu/scientific-portfolios_en

ensure their conservation by diminishing their physical manipulation, and could even preserve their memory in times of crisis or as a result of their destruction.

11 Conclusions and forward look

Gigapixel imaging sits at the intersection of **optics, electronics, and computer science**. It pushes each of those domains: lenses must be better, sensors more numerous or higher density, and computations more powerful. Over time, clever engineering solutions have emerged: multi-element lenses to flatten fields, arrays of sensors to sidestep single-sensor limits, automated stitching algorithms to handle alignment, and cloud platforms to handle distribution. The remaining challenges are being whittled away as technology improves.

As of now, one can reliably capture a multi-gigapixel panorama with off-the-shelf gear but pushing toward real-time gigapixel video or turnkey gigapixel cameras for consumers still requires further innovation (particularly in data handling and processing power). Nevertheless, the challenges that have been solved so far – from dealing with lens aberrations to stitching hundreds of thousands of images – showcase the impressive engineering that underpins gigapixel imaging. It's this union of precise hardware and advanced software that makes it possible to freeze a moment in time with breathtaking, billion-pixel clarity.

In practice, the mosaic approach remains the most popular for gigapixel photography in fields like tourism, sports, cultural heritage and virtual tours, due to the availability of robotic mounts and software. Exotic one-shot solutions (multi-sensor or custom cameras) are mostly found in research, military, or scientific contexts where real-time gigapixel capture is needed. Each method has trade-offs: mosaics are accessible but require time and stitching computation, whereas custom large-format or array cameras capture instantly but involve complex hardware and expense.

Gigapixel technology is reshaping how we observe, analyse, and interact with the world. From safeguarding communities against geohazards to protecting endangered species and exploring the cosmos, its applications are as diverse as they are impactful. As AI and hardware innovations continue to evolve, the future promises even more transformative uses, underscoring the power of high-resolution imaging to address global challenges.

Through gathering ideas from exhibition visitors and reviewing existing applications of this technology, a wide array of potential uses emerged across science, industry, and education. From the Disaster Risk Management perspective, in particular, potential experimental applications to geohazards assessments, environmental monitoring, industrial and emergency management, citywide surveillance and public safety, securing critical infrastructures could be explored, especially in the context of the envisaged JRC Research and Innovation Security Campus and the connected network of JRC relevant laboratories.

The Art@ECML exploratory project also served as a platform for experimenting with how tactile experiences and digital storytelling can enhance scientific communication and foster interactivity. Through interactive touchscreens (e.g., the ECML video wall), dynamic surfaces, and apps, users can engage with content in a personalised and immersive way. This approach could serve as a model for developing permanent installations at the newly opened Europe Science Experience (ESE) visitor centre of the JRC. This is even more relevant considering the "whole of society" approach to crisis management, as recommended in the European Preparedness Union Strategy[27] published in March 2025.

[27] https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52025JC0130

Finally, the Art@ECML exploratory project demonstrated the power of unconventional thinking, achieving remarkable success in two key areas: enhancing the visibility of the European Crisis Management Laboratory (ECML) among JRC colleagues and external Commission delegations, and positively impacting the wellbeing of JRC staff. By fostering informal networking in a creative, non-traditional setting, the project engaged the imaginative and emotional dimensions of scientists, offering a refreshing approach to workplace interaction.

References

ABC News. (2012, June 22). *AWARE2 Gigapixel Camera: 50,000 Megapixels and a Ton of Detail*. Retrieved from https://abcnews.go.com/Technology/aware2-gigapixel-camera-thousands-megapixels-lot-detail/story?id=16645484 abcnews.go.com

Brady, David J., Daniel B. Marks, David S. Kittle, David J. Schulz, Joseph M. Nothum, David R. Golish, Kenneth J. Webb, et al. "Multiscale Gigapixel Photography." *Nature* 486, no. 7403 (2012): 386–389. https://doi.org/10.1038/nature11150.

Designboom. (2012, June 27). *AWARE2 Gigapixel Camera*. Retrieved from https://www.designboom.com/technology/aware2-gigapixel-camera/ Designboom

Distant Focus Corporation. (n.d.). *AWARE Gigapixel Imaging*. Retrieved from https://distantfocus.com/projects/aware/ Designboom+3Distant Focus+3DukeSpace+3

Duke Information Spaces Project. (n.d.). *AWARE2 Multiscale Gigapixel Camera*. Retrieved from https://disp.duke.edu/research/aware2-multiscale-gigapixel-camera Duke Dispensary+1Duke Dispensary+1

Flint, G., & Aves, C. (2007). *The Gigapxl™ Project*. Retrieved from https://research.adobe.com/lecture/graham-flint-catherine-aves-gigapxl-project/ sandiegoreader.com+2Adobe Research+2SCPV+2

GigaPan Systems. "About Us." GigaPan. Accessed March 7, 2025. https://www.gigapan.com/cms/about-us.

HuffPost Tech. (2012, June 22). *Aware-2, New Gigapixel Camera, Takes Photos 30 Times Better Than Standard Cameras*. Retrieved from https://www.huffpost.com/entry/aware-2-new-gigapixel-camera-gigapan_n_1618879 HuffPost+1Byte Chronicles+1

Kurzweil Library. (2012, June 21). *A 50 Gigapixel Camera Five Times Better Than 20/20 Human Vision*. Retrieved from https://www.thekurzweillibrary.com/a-50-gigapixel-camera-five-times-better-than-2020-human-vision thekurzweillibrary.com

Rester M, Judmaier P, De Groeve T, Annunziato A, editors. Collaborative Human-Computer Interaction with Big Wall Displays - BigWallHCI 2013 3rd JRC ECML Crisis Management Technology Workshop. EUR 26396. https://publications.jrc.ec.europa.eu/repository/handle/JRC85533

Son, H. S. (2013). *Optomechanical System Development of the AWARE Gigapixel Scale Camera* (Doctoral dissertation, Duke University). Retrieved from https://dukespace.lib.duke.edu/items/590d8316-f07e-4310-9e70-288f9c0c3628 DukeSpace+1DukeSpace+1

Wired. (2005, February 15). *Photographer Seeks Resolution*. Retrieved from https://www.wired.com/2005/02/photographer-seeks-resolution/ Wired+1Wikipedia+1

List of abbreviations

Abbreviations	Definitions
AI	Artificial Intelligence
CCD	Charge-coupled device
CMS	Content Management System
DGTES	Digital Techno-Economic Ecosystem
DSLR	Digital Single-Lens Reflex
ECML	European Crisis Management Laboratory
ERCC	Emergency Response Coordination Centre
EU	European Union
FPGA	Field-Programmable Gate Array
GHSL	Global Human Settlement Layer
GIS	Geographic Information System
GPU	Graphics Processing Unit
IIIF	International Image Interoperability Framework
ISO	International Organization for Standardization
JRC	Joint Research Centre
LSST	Legacy Survey of Space and Time
MR	Mixed Reality
MuSCLE	Multi Source Crisis Layout Engine
OS	Operating System
SSD	Solid-State Drive
VR	Virtual Reality
WSI	Whole Slide Imaging X-ray
USB	Universal Serial Bus

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Annexes

Annex 1 – Art labels for the selected art pieces of the Art{@|×}ECML exhibition





ART{@|×}ECML

Technology to reconcile art and science



Adoration of the Magi (**Adorazione dei Magi**) c.1896
Gaetano Previati, born as (nato) Gaetano Cesare Previati, 1852
Oil on canvas (Olio su tela)
98 × 198 cm (38.6 × 78 in) - Pinacoteca di Brera, Milan (Milano)
References (Bibliografia)
Documentation by Pinacoteca di Brera
https://pinacotecabrera.org/en/collezione-online/opere/adorazione-dei-magi-2/

EN

Gaetano Previati's "Adoration of the Magi", painted in 1896, represents a departure from traditional depictions of this biblical event, embodying the spirit of Divisionism – a technique akin to Pointillism, characterized by the use of separate brush strokes of pure colour. This large canvas is a testament to Previati's innovative approach to light and colour, reflecting the artist's engagement with the Symbolist movement.

The Magi are seen in an almost ethereal setting, their figures dissolving into a whirl of vibrant colours and dynamic brushwork. The dramatic use of light creates a sense of spiritual radiance that surrounds the holy family. The Magi themselves are portrayed in a moment of humble reverence, offering their precious gifts with an air of solemnity and grace.

The scene unfolds in the vicinity of a small building, barely hinted at in its architectural details, as if to let the spectator's imagination complete the context. An indistinct crowd stands out in the background, where two camels emerge as iconic elements of the Magi's journey.

The painting invites contemplation, drawing the viewer into a mystical experience that transcends the physicality of the scene. Previati's work is not just a biblical representation but a poetic interpretation, where the play of light and colour seeks to evoke an emotional and contemplative response.

IT

L'"Adorazione dei Magi" di Gaetano Previati, dipinta nel 1893, rappresenta una svolta dalle rappresentazioni tradizionali di questo evento biblico, incarnando lo spirito del Divisionismo – una tecnica simile al Puntinismo, caratterizzata dall'uso di pennellate separate di colore puro. Questa grande tela è una testimonianza dell'approccio innovativo di Previati alla luce e al colore, riflettendo l'impegno dell'artista con il movimento Simbolista.

I Magi appaiono in un contesto quasi etereo, le loro figure si dissolvono in un vortice di colori vivaci e pennellate dinamiche. L'uso drammatico della luce crea un senso di radiosità spirituale che circonda la sacra famiglia. I Magi sono ritratti in un momento di umile riverenza, offrendo i loro preziosi doni con un'aria di solennità ed eleganza.

La scena si sviluppa nei pressi di una piccola costruzione, appena accennata nei dettagli architettonici, quasi a lasciare che sia l'immaginazione dello spettatore a completare il contesto. Sullo sfondo si staglia una folla indistinta, dove due cammelli emergono come elementi iconici del viaggio dei Magi.

Il dipinto invita alla contemplazione, attirando lo spettatore in un'esperienza mistica che trascende la fisicità della scena. L'opera di Previati non è solo una rappresentazione biblica, ma un'interpretazione poetica, dove il gioco di luce e colore cerca di evocare una risposta emotiva e contemplativa.

With the kind support of HALTADEFINIZIONE

ART{@|×}ECML

Technology to reconcile art and science



The Marriage of the Virgin (**Lo Sposalizio della Vergine**) c.1504
Raffaello Sanzio da Urbino 1483
Oil on panel (Olio su tavola)
170 × 118 cm (67 × 46.5 in) - Pinacoteca di Brera, Milan (Milano)
References (Bibliografia)
Documentation by Brera + Humanities
https://brera.in/humanities.it/opere/raffello-sposalizio-della-vergine/

EN

"The Marriage of the Virgin", completed around 1504 by Raphael Sanzio, is an exquisite representation of the High Renaissance style. The painting captures the moment of the marriage ceremony between the Virgin Mary and Saint Joseph, set against the backdrop of an elegantly designed temple, which demonstrates Raphael's mastery of perspective and architectural detail.

Raphael's composition is balanced and harmonious, with figures arranged in a semicircle, echoing the shape of the temple's grand portico. The central figures of Mary and Joseph are depicted with a serene dignity, as Joseph places the wedding ring on Mary's finger, symbolizing their divine union. The other suitors, busy breaking their dry rods, stand as testament to the miraculous nature of Joseph's flowering rod and thus selection as Mary's spouse, as told by the Protoevangelium of James.

The painting's vibrant colours and clear light contribute to the tranquil and sacred atmosphere, while the detailed expressions and gestures of the figures convey the solemnity of the occasion. Raphael's work reflects the ideals of the High Renaissance – clarity, order, and beauty – and it remains a powerful example of narrative storytelling through art.

In 2020, for the five-hundredth anniversary of Raphael's death, Haltadefinizione digitized and faithfully reproduced "The Marriage of the Virgin". Thanks to this ambitious initiative to relocate artistic heritage, after 222 years, the Church of San Francesco in Città di Castello, near Perugia, has regained the original pictorial harmony and colours of this work. One of Raphael's early works, it remains among the most celebrated of the Urbino-born painter, who completed it at age 21 in 1504 for this very Città di Castello, from which it was later removed in 1798 at the hands of Napoleon's General Lechi.

A three-dimensional replica of the panel, now housed in the Pinacoteca di Brera, was obtained with advanced 3D capture and printing technologies, faithfully reproducing details such as brushstrokes, cracks, and imperfections. The digitization, based on 4250 frames processed in 15 days, generated a 1.13 TB file, allowing a physical and chromatic replica identical to the original.

IT

"Lo Sposalizio della Vergine", completato intorno al 1504 da Raffaello Sanzio, è una rappresentazione squisita dello stile dell'Alto Rinascimento. Il dipinto cattura il momento della cerimonia nuziale tra la Vergine Maria e San Giuseppe, impostato sullo sfondo di un tempio elegantemente progettato, che dimostra la padronanza di Raffaello nella prospettiva e nel dettaglio architettonico.

La composizione di Raffaello è equilibrata e armoniosa, con le figure disposte in un semicerchio, che fa eco alla forma del grande portico del tempio. Le figure centrali di Maria e Giuseppe sono raffigurate con una serena dignità, mentre Giuseppe infila l'anello nuziale al dito di Maria, simbolo giando la loro unione divina. Gli altri pretendenti, impegnati a spezzare le loro verghe secche, testimoniano la natura miracolosa della verga fiorita di Giuseppe e quindi la sua scelta come sposo di Maria, come racconta il Protovangelo di Giacomo.

I colori vibranti e la luce chiara del dipinto contribuiscono all'atmosfera tranquilla e sacra, mentre le espressioni dettagliate e i gesti delle figure trasmettono la solennità dell'occasione. L'opera di Raffaello riflette gli ideali dell'Alto Rinascimento – chiarezza, ordine e bellezza – rimanendo un potente esempio di narrazione attraverso l'arte.

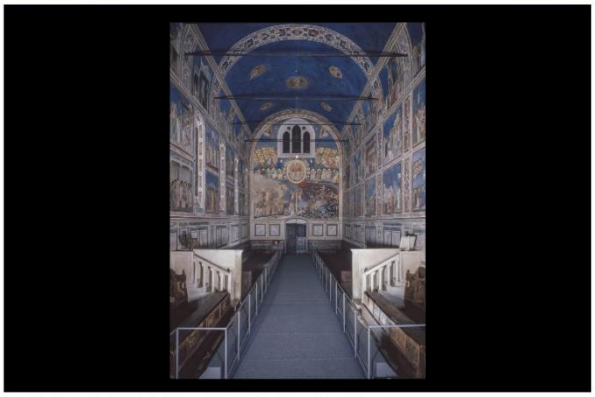
Nel 2020, per il cinquecentenario dalla morte di Raffaello, Haltadefinizione ha digitalizzato e riprodotto fedelmente "Lo Sposalizio della Vergine". Grazie a questa ambiziosa iniziativa di ricollocazione del patrimonio artistico, dopo 222 anni, la Chiesa di San Francesco a Città di Castello (PG) è tornata ad accogliere l'armonia pittorica e le cromie originarie di quest'opera. Opera giovanile di Raffaello, rimane fra le più celebrate dell'Urbinate, che la ultimò ventunenne nel 1504 proprio per questo Città di Castello, da cui venne poi rimossa nel 1798 per mano del generale napoleonico Lechi.

Una replica tridimensionale della tavola, oggi conservata nella Pinacoteca di Brera, è stata ottenuta con tecnologie avanzate di acquisizione e stampa 3D, riproducendo fedelmente dettagli come pennellate, cretti e imperfezioni. La digitalizzazione, basata su 4250 fotogrammi elaborati in 15 giorni, ha generato un file di 1,13 TB, permettendo una replica fisica e cromatica identica all'originale.

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ART{@|×}ECML

Technology to reconcile art and science



Scrovegni Chapel Frescoes (Affreschi della Cappella degli Scrovegni) 1303-1305 Giotto di Bondone 1267 Fresco (Affresco)
Chapel nave (navella della cappella): 29.88 × 12.65 × 8.41 m (98 × 41.5 × 27.6 ft) - Scrovegni Chapel (Cappella degli Scrovegni), Padua (Padova)
References (Bibliografia)
Documentation by Padova Musei Civici: https://www.cappelladegliscrovegni.it/index.php/it/la-cappella-di-giotto
Documentation by the Municipality of Padua: https://www.padovanet.it/informazione/cappella-degli-scrovegni
Documentation by UNESCO: https://whc.unesco.org/en/list/1625
A. Verdi, L'architettura della Cappella degli Scrovegni, in Giotto e il suo tempo, Motta, Milan 2000, pp. 118-158.

EN

The Scrovegni Chapel, adorned with Giotto di Bondone's frescoes, is a masterpiece of Western art and a cornerstone of the Italian Renaissance. Completed around 1305 in Padua, the fresco cycle narrates the life of the Virgin Mary and the life and passion of Christ in a series of 38 panels.

Giotto revolutionized the art of painting with his use of space, volume, and perspective, bringing unprecedented realism and emotion to the biblical stories. Each scene is a vivid tableau, filled with human expression and dramatic intensity. The frescoes offer a cohesive visual narrative, leading from the Joachim and Anna stories at the top to the Passion of Christ and culminating in the Last Judgement on the entrance wall.

The azure sky, which serves as a backdrop for many of the panels, gives the chapel an open, heavenly feel, while Giotto's skilful rendering of light and shadow imparts depth and solidity to the figures. His work here is a testament to his understanding of human emotion, gesture, and interaction, setting the stage for future artistic developments.

In 2009, the entire cycle of frescoes was acquired by Haltadefinizione with gigapixel technology under a concession from the City of Padua. In order to survey the entire painted surface of more than 700 square meters, a massive photographic campaign was carried out for a total of 14,000 shots and a year of work to process the images.

As you explore the digital representation of the Scrovegni Chapel, immerse yourselves in Giotto's world, examining the intricate details and the expressiveness of each fresco. The interactive experience allows for an appreciation of the subtleties that make this cycle one of the most important contributions to the history of art, which in 2021 lead to UNESCO declaring Padua's fresco cycles a World Heritage site.

IT

La Cappella degli Scrovegni, ornata dagli affreschi di Giotto di Bondone, è un capolavoro dell'arte occidentale e una pietra miliare del Rinascimento italiano. Completato intorno al 1305 a Padova, il ciclo di affreschi narra la vita della Vergine Maria e la vita e la passione di Cristo in una serie di 38 pannelli.

Giotto rivoluzionò l'arte della pittura con il suo uso dello spazio, del volume e della prospettiva, donando alle storie bibliche un realismo e un'emozione senza precedenti. Ogni scena è una rappresentazione vivida, colma di espressione umana e intensità drammatica. Gli affreschi offrono una narrativa visiva coerente, che va dalle storie di Gioacchino e Anna in alto fino alla Passione di Cristo e culmina nel Giudizio Universale sulla parete d'ingresso.

Il cielo azzurro, che funge da sfondo per molti dei pannelli, conferisce alla cappella un'atmosfera aperta e celestiale, mentre la sapiente resa di luce e ombra di Giotto conferisce profondità e solidità alle figure. Il suo lavoro qui è una testimonianza della sua comprensione dell'emozione umana, del gesto e dell'interazione, che prepara il terreno per futuri sviluppi artistici.

Nel 2009, l'intero ciclo di affreschi è stato acquisito da Haltadefinizione con tecnologia gigapixel su concessione del Comune di Padova. Per rilevare l'intera superficie dipinta di oltre 700 metri quadrati, è stata realizzata una imponente campagna fotografica per un totale di 14.000 scatti e un anno di lavoro per l'elaborazione delle immagini.

Mentre esplorate la rappresentazione digitale della Cappella degli Scrovegni, immergetevi nel mondo di Giotto, esaminando i dettagli intricati e l'espressività di ogni affresco. L'esperienza interattiva permette di apprezzare le sottigliezze che rendono questo ciclo uno dei contributi più importanti nella storia dell'arte e che ha portato l'UNESCO a dichiarare i cicli di affreschi padovani Patrimonio dell'Umanità.

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Annex 2 – Postcards on details of the art pieces selected for the Art{@|×}ECML exhibition, and links to various scientific/artistic domains

Art{@|×}ECML introductory postcard



"Primavera" painting of Botticelli and link with Cultural heritage preservation domain



"Primavera" painting of Botticelli and link with the Biodiversity/flora domain



“Adoration of the Magi” painting of Gentile da Fabriano and link with Biodiversity in fauna



“Adoration of the Magi” painting of Gentile da Fabriano and link with Landscape and urban planning



“Adoration of the Magi” fresco of Giotto and link with Astronomy



“The marriage of the Virgin” painting of Raffaello Sanzio and link with Mathematics/ architectural perspective



"Adoration of the Magi" painting of Previati and detail on the Symbolism/divisionism painting technique



"Primavera" painting of Botticelli and the Artist's rethinking



"The marriage of the Virgin" painting of Raffaello Sanzio and the symbolism of rings





'The Marriage of the Virgin', completed around 1504 by Raffaello Sanzio, captures the moment of the marriage ceremony between the Virgin Mary and Saint Joseph. The other suitors, busy breaking their dry rods, stand as testament to the **miraculous nature of Joseph's flowering rod** and thus selection as Mary's spouse, as told by the **Protoevangelium of James**.

Raffaello masterfully uses the **language of rings**, painting them to highlight the social affiliation of the characters, their physical characteristics, and their intellectual and moral qualities. In this case, he does not show us rubies or sapphires because **Maria's ring** serves as a **symbol that makes sacred the promise** of union, i.e. the marriage of the Virgin.

What ideas does this example inspire in you?
How could we adapt this technology to the visual data and analyses you work with?
Use the QR code to share your thoughts and help us take this experiment further.

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Annex 3 - Comments and suggestions received from the visitors to the exhibition

- *At the **Global Human Settlement Layer** (GHSL), it would be great to visualize our maps on the ECML screen.*
- *This technology could be applied in cities for **urban warfare planning, mitigation or operational in a crisis**. Supporting layers in a GIS of infrastructure, information etc would support.*
- *It would be interesting to see this applies the field of **digital pathology and cellular research**. Images from the specimen can be combined with microscopic images of tissue sections for pathology applications in healthcare teaching. This can be further combined with spatial omic data for research to better map and understand the evolution of a disease namely for cancer but also other diseases.*
- *Could make a visual representation of the **Digital Techno-Economic Ecosystem** (DGTES) network (firm network in the digital sphere!)*
- ***Biometric identification** from distant cameras*
- *I assume you are already in contact with the **sci art team and with digital heritage** people in RTD*
- *Not in my work, however very high-resolution imagery might be applied to **identify sub-millimeter details and anomalies in the surface of objects, for instance reactor tanks**, that cannot be analysed with other means (microscope, UV, X-ray...). The technology, a sophisticated version of Google's Street View, is extremely storage hungry and as such not particularly eco-friendly. I do currently not see this applied in other fields, for instance landscaping technologies where means like Lidars already allow for sufficient resolution, or space imagery. In my opinion the resolution of current smartphone cameras or SLRs is sufficient to cope with almost any practical need. Do also not see Hubble-type satellites -which have comparable resolution- observing the finest details on Earth and being concurrently also GDPR compliant.*
- *"In transport, high-resolution imagery can help in **monitoring infrastructure conditions**, such as road surface wear, bridge integrity, and other critical components that may require maintenance. Drones equipped with high-resolution imaging can create detailed topographical maps, essential for planning transportation routes and assessing land for development.*
- *In **border security**, drones with high-resolution cameras can provide real-time surveillance, allowing for better situational awareness and faster response to potential threats. Moreover, **monitoring traffic flow and congestion** through high-resolution images can lead to more effective management of transportation systems."*
- *Fantastic for **land management and carbon dynamics***
- *Many ways - as project leader of **art and science** I would recommend presenting this at our visiting artists in near certainty that some of them will be enthusiastic you experiment with this technology*
- *Use it to display **spatial (geographical) data***
- *For increasing more **public engagement and awareness**, especially to the youth and children, our COPE disaster initiative books is applying this technology - leveraging visual storytelling to increase their DRR knowledge in order to become more resilient.*
- *Would be nice to use it to see the **fractal nature of natural systems***
- *Include **dynamic images** (eg flows in the ocean or atmosphere) like NASA's hyperwall*
- *Congrats! Suggest to add **pictures taken in infrared***
- *Useful to put on the **welcome center with one picture for each jrc site***
- *Thank you for sharing this with us all! It would be also good to see the crisis room displayed how it is used currently.*
- *Thank you very much for the experience, it was very educating! Also, I got as a bonus an overview of the ECML and it was super interesting!*
- *Please check out David Monacchi and his **Sonosfera***
- *By **blending the emotional depth of art with the innovative power of technology**, crisis management can be made more human-centered, impactful, and adaptable. Each field can enhance the others: art can humanize and deepen technological responses, while technology can amplify and expand the reach and impact of artistic efforts.*
- *Making daily life better. I want one at home*

Annex 4 - Cameras and Imaging Systems Capable of Gigapixel Imaging

Achieving gigapixel imagery demands specialized camera systems or setups that can either capture or compile extremely high-resolution data. Below we examine notable cameras and imaging instruments known for gigapixel capabilities, along with their optical and technical designs:

- **GigaPan Panoramic Systems:** The GigaPan device series (EPIC, EPIC 100, EPIC Pro) is a family of robotic camera mounts specifically designed to enable gigapixel photography with standard cameras. Developed from technology used on NASA's Mars rovers and a collaboration between Carnegie Mellon University and NASA/Google in 2008 (phys.org), GigaPan mounts automate the process of capturing multiple overlapping photos for a panorama. The optical component in a GigaPan setup is simply the user's camera and lens – often a DSLR with a telephoto lens for maximum detail. The mount precisely rotates and tilts the camera through a programmed grid of shooting positions, while triggering the shutter for each exposure. Key design considerations include robust motors and gearing to support heavy camera+lens combos (the EPIC Pro supports up to ~10 lbs) and adjustable brackets to align the camera's nodal point, minimizing parallax (newatlas.com, currentphotographer.com). Using a GigaPan is straightforward: the photographer sets the panorama's corners in the unit, and the device calculates how many photos (often hundreds or thousands) are needed and in what pattern (newatlas.com). It then proceeds to capture each tile in sequence with precise overlaps. After shooting, images are fed into stitching software (GigaPan Stitch or similar) which automatically aligns and blends them into a single gigapixel image (newatlas.com). The GigaPan EPIC series and similar pano-heads essentially turn a normal camera into a gigapixel imaging system by adding electromechanical precision and companion software. Optically, the quality depends on the camera's sensor and lens; for example, using a 20 MP DSLR with a long focal length lens might require ~50–100 frames for a gigapixel, whereas a 50 MP camera could achieve the same with fewer frames. GigaPan systems have been used widely: from crowds at sporting events to Obama's 2009 inauguration, creating interactive images where viewers can zoom in to "find themselves" in the scene (phys.org, bhphotovideo.com). The combination of commodity cameras with specialized robotics and stitching algorithms exemplifies how gigapixel imaging became accessible.

- **Custom Gigapixel Cameras (Multi-Lens Systems):** Purpose-built gigapixel cameras like the **AWARE series** push the envelope of optical design. The AWARE-2, developed by Duke University and Distant Focus Corp., is a *multi-scale* camera consisting of a large spherical primary lens that projects an image onto an array of 98 micro-cameras, each with its own lens and sensor (petapixel.com). In essence, it's like 98 cameras looking through one eye. The optical design is *monocentric*: a ball-shaped lens that affords a wide field (120°) with uniform imaging onto a curved focal surface, which is then partitioned among the many 14 MP sensors. The AWARE-2's electronics synchronize these micro-cameras so that they expose simultaneously and then feed the image data to a central processor. The stitching in this case is computational fusion of the sub-images, since each module knows its fixed place in the mosaic (so it's more like assembling puzzle pieces than finding overlaps). The result is a single 50 gigapixel composite with five times the resolution of 20/20 human vision (sciencedaily.com. Notably, **97% of the AWARE-2's volume is taken up by electronic boards, wiring, and cooling**, with only 3% for optics (petapixel.com), underscoring the data-processing challenge of so many sensors. A later iteration, AWARE-10, increased the sensor count (382 micro-cameras) and achieved nearly 3 gigapixels per exposure, while improving compactness and efficiency (distantfocus.com). These cameras require meticulous optical alignment – *thousands* of individual optical elements (lenses, sensors, etc.) must be positioned within microns to achieve the intended resolution (spie.org). The payoff is the ability to capture dynamic scenes at gigapixel detail in real time (useful for surveillance or wide-area video). Outside of Duke's projects, others have explored multi-sensor gigapixel cameras e.g., a Microsoft Research prototype with 100 micro-cameras, the DARPA ARGUS system, but none are commercially common yet due to their complexity.

- **Gigapixel Microscope Imaging Setups:** In biomedical and scientific imaging, gigapixel scales are reached by combining microscopy with panorama techniques. **Whole Slide Imaging (WSI)** canners, for example, digitize pathology microscope slides at high magnification. These devices use a microscope objective (20× or 40×) and a high-resolution sensor to scan the slide in tiles. A motorized stage moves the slide in a grid, capturing thousands

of adjacent fields that are stitched into one enormous image sometimes exceeding 100,000×100,000 pixels (tens of gigapixels) (arxiv.org, osti.gov). The optical design is essentially a microscope (often with telecentric lenses to minimize distortion across the field) combined with precise stage control. Focus stacking may be employed to ensure each tile is sharp. The resulting gigapixel images allow pathologists to examine tissue samples digitally with the same detail as a real microscope. Technical documentation notes that a single whole-slide image can require several gigabytes of storage and memory to handle (osti.gov), necessitating special image formats and viewers. Another approach in microscopy is using **multi-camera array microscopes**. Researchers have built systems with dozens of mini cameras pointed at a specimen to achieve a wide field-of-view at cellular resolution in one composite shot (horstmeyer.pratt.duke.edu). For instance, a 96-camera array microscope can capture a 20 cm diameter area in one go, producing a gigapixel-scale image (or even video) of, say, a large biological sample with microscopic detail (horstmeyer.pratt.duke.edu). Each module covers a subsection of the sample, and software stitches the sub-images in real time. These complex instruments are advancing neuroscience and cytology by recording activity over large areas without sacrificing detail. In summary, whether by *scanning small fields* or *parallelizing many small imagers*, microscope-based gigapixel systems extend the concept of "billion-pixel photography" to the micro-world.

- **Notable Large-Scale Cameras:** In astronomy, the **LSST (Legacy Survey of Space and Time) Camera** deserves mention as an imaging system of gigapixel order. It is essentially a gigantic digital camera attached to a telescope, with a focal plane of 189 CCD sensors totalling 3.2 gigapixels (lsst.org). The sensors are deep-cooled and read out in parallel, capturing a 9.6-square-degree patch of sky in each exposure. Similarly, the earlier Pan-STARRS and other survey telescopes employed mosaic CCD cameras (Pan-STARRS had 1.4 gigapixels) to map the sky. While these are not "cameras" in the handheld sense, they are engineered imaging systems that confirm the gigapixel frontier is not limited to panoramas – it spans scientific instruments as well.

Optical and Sensor Technologies: Across these examples, a few technical themes appear. High-end gigapixel systems often use *monocentric lenses* or multi-element wide-angle optics to project a large, sharp image circle (Duke's cameras and some microscope designs use such lenses) (sciencedaily.com). Sensor technology ranges from standard CMOS/CCD chips tiled together to scanning mechanisms; there is no single "gigapixel sensor" chip commercially available (the largest single sensors today are on the order of tens of megapixels), so combining multiple sensors or multiple exposures is necessary. Image stitching software is a critical component of many systems – whether it's a program like GigaPan Stitch/Equirectangular projection for panoramas or a custom GPU pipeline for fusing micro-camera data (sciencedaily.com). Many gigapixel rigs also incorporate precise motion control (robotic pan heads, translation stages, etc.) to ensure overlap and consistent imaging geometry. Data handling is non-trivial too: gigapixel cameras often require high-speed data buses or RAID storage to ingest the flood of pixels, and in some cases onboard processing to compress or reduce the data in real time. We will explore some of these engineering challenges in a later section. But as a survey, the current landscape of gigapixel imaging spans from relatively simple **panoramic setups** (camera + robotic head + stitching software) to **cutting-edge multi-sensor behemoths** with custom optics. Each is a solution to the same problem: how to collect an immense number of pixels from a scene to create an ultra-high-res image.

Annex 5 - Historical Evolution of Gigapixel Imaging

Gigapixel imaging, as a field, has evolved dramatically over the past two decades. Early efforts were often experimental or niche, but rapid advances in digital imaging and computation have turned gigapixel photography from a rarity into a more routine endeavour. Below is an outline of key milestones and technological advancements over time.

Early 2000s – Pioneering Experiments: The concept of a "gigapixel photo" emerged around the turn of the millennium. The **Gigapxl Project** (2000–2004) spearheaded by Graham Flint was a seminal effort to break the gigapixel barrier. Flint, a retired physicist, constructed a one-of-a-kind large format camera using Cold War-era optics and a massive film back (wired.com). By 2003, he succeeded in capturing and scanning the first digital images exceeding one billion pixels. Photographer Max Lyons is credited with creating *the first stitched gigapixel image* in 2003, which he exhibited in Las Vegas in 2004 (tawbaware.com). Lyons achieved this by stitching many digital photos (shot with a ~6 MP DSLR of the time) into a panorama, pushing the limits of then-available software and hardware. These early gigapixel images were groundbreaking but *difficult to produce*, often requiring custom tools or painstaking manual work. In Flint's case, he had to design custom lenses and use drum scanners to digitize film; in Lyons' case, software like Panorama Tools and enormous computing power (for that era) were needed to process the data. The results, however, were awe-inspiring – Flint's 4-gigapixel landscape prints astounded viewers with detail (e.g. distant people or text visible in a wide vista) and proved that **ultrahigh resolution photography** was achievable (sandiegoreader.com, wired.com).

Mid-2000s – Advancements in Stitching and Hardware: The mid-2000s saw rapid improvement in the tools for gigapixel imaging. On the software side, algorithms for image stitching and alignment improved significantly – notably the introduction of **Autostitch (2007)**, which used SIFT feature matching to automatically align panoramas, and commercial programs like PTGui and Adobe Photoshop's Photomerge. These removed much of the manual labor in creating mosaics. Camera sensor resolutions were also climbing (from ~6 MP around 2002 to ~12–16 MP by 2006 in high-end DSLRs), which meant fewer images were needed to reach a gigapixel. However, perhaps the biggest catalyst was the development of **robotic pano heads** and integrated systems. In 2006–2008, Carnegie Mellon University and NASA Ames (with Google's support) refined the GigaPan prototype and launched it commercially in 2008 (gigapan.com). By 2010, the GigaPan EPIC series became widely available, allowing enthusiasts to relatively easily shoot gigapixel panoramas (currentphotographer.com). This democratized gigapixel imaging – one no longer needed a one-of-a-kind rig; a prosumer camera and a $900 robot mount could do the job. As a result, photographers around the world started attempting record-breaking panoramas. We saw gigapixel photos of city skylines, like a 16 GP image of Dresden in 2009, a 26 GP image of Paris in 2010, and a 45 GP image of Dubai. Each year seemed to bring a new "world's largest photo." By 2012, a panorama of London taken from the BT Tower reached **320 gigapixels**, a staggering number that held the record at the time. The trend was clear: improved *computing power* and *software* could handle more images, and improved *camera sensors* and *optics* captured more detail per shot, compounding to push gigapixel counts ever higher. As Max Lyons noted reflecting on the 2000s, creating gigapixel images went from "extremely difficult" to gradually easier with *"improved commercial software and robotic equipment,"* such that many "record-seekers" jumped in by the end of the decade (tawbaware.com). In short, what began as a rare technical feat became almost mainstream in the panorama community.

2010s – Gigapixel Goes Mainstream and Multiscale: In the 2010s, gigapixel imaging solidified in multiple domains. On the consumer side, projects like the **Gigapixel FanCam** appeared at sporting events and concerts, allowing fans to explore crowds in extreme detail online (the technology was essentially GigaPan or similar pano rigs paired with web viewers) (phys.org). Gigapixel panoramas became common in marketing and journalism – for example, sports finals, large festivals, and city tourism boards began commissioning such images. The quality also improved thanks to better stitching algorithms that could correct exposure differences and minimize seam artifacts. By mid-2010s, even *multi-row panoramas of moving crowds* could be stitched with few errors (though ghosting still occurs, like the "missing arms" when people move between shots (phys.org). The availability of online platforms (GigaPan.org, Microsoft's HDView, Zoomify, etc.) made it easy to share and explore these giant images.

Meanwhile, in scientific/industrial realms, **multi-scale gigapixel imaging** was advancing. A landmark development was the aforementioned **AWARE-2 camera**, published in *Nature* in 2012 (nature.com). This demonstrated that you could capture on the order of 10–50 gigapixels *in a single exposure* using a novel camera architecture. It was a proof-of-concept that tackled the fundamental limits of traditional cameras (where increasing resolution is constrained by lens size and sensor size) by breaking the problem into pieces – literally using many small lenses/sensors. The AWARE project's success suggested a possible future where gigapixel cameras might become portable and common, once electronics could be miniaturized and cost-reduced (sciencedaily.com). Throughout the 2010s, that vision inched closer: the AWARE-10 (2013) increased resolution further, and DARPA explored even larger formats for wide-area surveillance (there were reports of a 1.8 gigapixel drone camera system called ARGUS). In microscopy, digitization of pathology slides went from experimental to routine in many labs, producing multi-gigapixel images daily. Google's **Art Camera** (launched 2016) applied gigapixel mosaic imaging to museum artworks globally, capturing hundreds of famous paintings in ultra-high detail (livescience.com, shutterbug.com). By the late 2010s, we also saw gigapixel imaging enter pop culture and the arts: for instance, the Rijksmuseum in Amsterdam released a 44.8 gigapixel image of Rembrandt's *Night Watch* in 2020, only to follow up in 2022 with an astounding **717 gigapixel** version created from thousands of close-ups (frontiersin.org). This set a record for the highest resolution image of an artwork, and it required not only advanced robotic capture but also massive computing resources to process (the raw data and final file size were in the multi-terabyte range).

2020s – Beyond Gigapixel (Terapixel Era): Entering the 2020s, the frontier has pushed into *terapixel* territory (trillions of pixels). While no single camera can snap a terapixel image outright, composite techniques have achieved this milestone. A notable example is the **world's first terapixel macro photo** created in 2018, which was a stitched image of a food mosaic featuring **1.053 terapixels (1053 gigapixels)**. It was assembled from 629,370 individual photographs taken over five days with multiple robotic rigs (petapixel.com) – essentially an extreme extrapolation of the mosaic approach. Microsoft's Terapixel Project (2010) also demonstrated a terapixel image of the night sky by combining imagery from sky surveys; they processed it on a supercomputer cluster and produced a seamless spherical sky image over 1 trillion pixels in size (microsoft.com). These feats underscore how far image stitching and processing have come. At the same time, *gigapixel hardware* continues to mature: the Vera Rubin Observatory's 3.2 gigapixel camera saw first light and is expected to generate 15 TB of image data each night once operational, ushering in an era of "big data" in imaging. Researchers are even discussing **petapixel** imaging theoretically – for example, arrays of cameras or future sensor tech that could hit 10^15 pixels for surveillance or space imaging (nextbigfuture.com). While a true petapixel image is yet to be made, the fact that gigapixel and terapixel images exist and are shared widely is impressive enough. The evolution from one-off gigapixel experiments to daily gigapixel captures (e.g. every panorama in Google Street View is hundreds of megapixels to gigapixels in composite) reflects advancements in *sensor resolution, computational photography, and computing power*. What once took a dedicated expert with custom gear can now be done by a skilled amateur with off-the-shelf tools – and that democratization is a hallmark of the technology's evolution.

In summary, gigapixel imaging has progressed from an ambitious idea to a practical technique used in fields ranging from landscape photography to microscopy and astronomy. Key advancements – **higher-resolution sensors**, **better optics, robotic mounts**, and above all **powerful software/processing** – have raised the achievable resolution from 1 GP to 1000+ GP in twenty years. Today, gigapixel images are not only more common but also higher quality (in terms of dynamic range, stitching accuracy, etc.) compared to early attempts. As technology continues, we can expect gigapixel imaging to further integrate with computational imaging (AI-assisted stitching and enhancement) and perhaps real-time video applications. The historical trend is clear: the ceiling on resolution keeps rising.

Annex 6 - Beyond Gigapixel: Related Imaging Technologies

Gigapixel imaging is part of a broader family of ultra-high-resolution imaging techniques. Briefly, some related technologies and concepts include:

- **Terapixel Imaging:** As noted, *terapixel* images (10^12 pixels) have been achieved via mosaicking. Microsoft's WorldWide Telescope incorporated a terapixel sky image, which involved stitching thousands of astronomical photographs and required significant computational resources (a 64-node cluster processed the data into an 802 GB image file) (vizworld.com). Terapixel-scale images remain composite endeavours, often for artistic or scientific showcases, since storing and manipulating a trillion-pixel image is extremely challenging (typically they are kept in pyramid-based tiled formats for viewing). The GIGAmacro project's terapixel mosaic in 2018 demonstrated that even at this scale, interactive exploration is feasible with the right viewer software (petapixel.com). These efforts foreshadow a time when trillion-pixel images might be more routine, especially if sensor arrays grow and computing enters the exascale regime.

- **Computational Panorama and Light-Field:** While not gigapixel in raw count, light-field cameras and synthetic aperture photography assemble data from multiple micro-cameras or viewpoints to enhance resolution or refocus images. For example, the Lytro camera uses an array to capture angular light information; one could imagine a hybrid of light-field and gigapixel (though current light-field cameras are low resolution). Similarly, synthetic aperture radar (SAR) in remote sensing combines signals from a moving radar to form an image with very high equivalent resolution – conceptually akin to stitching, but with coherent signals. These techniques relate to gigapixel imaging in that they involve merging multiple samples to exceed single-sensor limits (trading one kind of complexity for another).

- **Multiscale "Terapixel" Cameras:** In defence and surveillance research, there has been discussion of eventually building **multi-gigapixel or terapixel camera systems**. The AWARE project, for instance, had a roadmap for scaling to 10 and 50 gigapixels by clustering more micro-cameras (nextbigfuture.com). Such a device, if realized, would effectively be a *"terapixel camera."* It would produce an enormous data stream (possibly requiring on-the-fly data reduction via AI). While no publicly known camera has reached a terapixel in one shot, the concept is an extension of the multi-sensor approach – it's plausible with enough parallel sensors and advanced optics (monocentric lenses can, in theory, scale up with more cameras tiling the focal sphere). The fundamental limits here involve physics (diffraction and aberrations across huge apertures) and practicality (power and data handling), so these remain experimental.

- **Extreme Gigapixel Visualization:** Another related area is how to *display or visualize* such massive imagery. Technologies like tiled wall displays, VR environments, or AR allow users to experience gigapixel data in context. For instance, a museum might use a large multi-screen installation to let visitors zoom into a gigapixel scan of a historic painting. Or VR headsets might let users "step inside" a 360° gigapixel panorama. These aren't cameras per se, but they are part of the ecosystem of working with ultra-high-res content.

In essence, while gigapixel usually denotes still images ~10^9 pixels, the offshoot and next-step technologies deal with orders of magnitude beyond (terapixel) or new ways of capturing that data (camera arrays, computational methods). The primary focus remains gigapixel imaging, but it is important to note that the drive for higher resolution continues. As one researcher quipped, ubiquitous gigapixel cameras may change photography from "where to point the camera" to "how to mine the data" (petapixel.com) – meaning once you capture everything in extreme detail, the challenge shifts to extracting useful information. This sentiment grows even more relevant when considering terapixel or continuous gigapixel video systems. Thus, gigapixel imaging can be seen as a bridge between conventional photography and the data-intensive imaging modalities of the future.

Annex 7 - Technical Considerations and Engineering Challenges in Gigapixel Imaging

Creating and working with gigapixel images presents significant technical hurdles. From capturing the raw data to processing and storing the final image, engineers and photographers must solve problems not encountered in normal photography. Here we discuss some key considerations, and the solutions developed to address them:

- **Optics and Lens Precision:** Achieving uniform, sharp focus across a gigapixel image is exceedingly challenging. If using a single lens to cover a wide field (as in large-format or multi-sensor cameras), the lens must be *exceptionally well-corrected* to avoid blurriness or aberrations at the periphery. Graham Flint's Gigapxl camera underscored this point – he found no off-the-shelf lens met his requirements, so he and an optical designer built the *Asymmagon*, an 8-element custom lens optimized for a 9″×18″ film frame (sandiegoreader.com). This lens eliminated chromatic aberration and maintained high resolution corner to corner, but took a year of design and testing (sandiegoreader.com). In multi-sensor array cameras like AWARE-2, the solution was a monocentric lens (spherical symmetry helps uniformity) paired with multiple small secondary lenses. Still, aligning these optics is a tour-de-force: in the AWARE-10 (4 GP) camera, *about two thousand optical components* had to be aligned to very high accuracy to achieve the desired resolution across the image (spie.org). Even slight misalignments could cause stitching seams or focus errors. Engineering approaches to this include precision machined lens mounts, active focus mechanisms (AWARE uses actuators to micro-adjust sensor positions for focus), and careful calibration. Another optical issue is *depth of field* – in mosaic imaging, if the scene isn't flat, different images may have different focus. Robotic pano heads often integrate focus stacking or recommend shooting at small apertures to maximize depth of field. In microscopy, automated focus stacking is used when scanning a slide (since tissue sections may not be perfectly flat). In summary, the optical engineering of gigapixel systems strives to maintain *consistency and sharpness* over large image planes, which often involves custom lenses or complex assemblies far beyond a normal camera's requirements.

- **Sensor and Sensitivity Considerations:** Gigapixel setups frequently rely on multiple smaller sensors rather than one giant sensor. Smaller pixels or many sensors can introduce issues of noise and sensitivity. For example, the microcameras in AWARE-2 were 14 MP each with 1.4 µm pixel size (elifesciences.org) – such small pixels can be less light-sensitive and more prone to noise, especially in low light. To compensate, designers use *large apertures* (the AWARE's primary lens was f/2.4, relatively fast for such a wide field) and may incorporate cooling or shorter exposure times to avoid overheating sensors. Additionally, when combining images from different sensors or different exposures, variations in sensor response must be corrected. Each sensor might have slightly different color characteristics or gain. A lot of *calibration* is needed so that the final mosaic doesn't show boundaries. Methods include capturing flat-field images to correct vignette and gain differences, and matching color balance between tiles. In panoramic mosaics with a single moving camera, the challenge is more about exposure differences (the sky might brighten during the capture sweep, etc.). Photographers mitigate this by using manual exposure or locking exposure for the whole set, and advanced stitching software can adjust intensity seams. Microsoft's Terapixel project, for instance, had to *"systematically remove differences in exposure, brightness, noise floor and color"* across the source photos to create a seamless result (news.microsoft.com). As for *dynamic range*, some gigapixel captures bracket exposures (the GigaPan can trigger multiple shots per position for HDR blending (newatlas.com) to handle high contrast scenes. In summary, maintaining consistent sensitivity and color across a gigapixel image demands careful sensor calibration and often post-processing equalization. Modern systems handle a lot of this with software – for example, stitching programs use photometric correction algorithms to even out brightness and tone across the panorama.

- **Data Volume and Storage:** The sheer amount of data in gigapixel imaging is a major engineering concern. A single uncompressed gigapixel image (10^9 pixels with 3 bytes per pixel in 8-bit RGB) is about 3 GB; in practice many gigapixel images are larger (due to higher bit depth or multiple gigapixels). For instance, a 4 gigapixel image would occupy roughly 24 GB uncompressed (engadget.com). Handling files of this size requires robust computing resources. In the early days, just opening or editing such files in standard image software was impossible. This led to development of specialized formats like BigTIFF (TIFF variant for >4 GB files) and multiresolution tiled formats – essentially the image is not treated as one huge bitmap in memory, but as tiles that can be loaded as needed.

Viewing a gigapixel image also needs a tiled, multiresolution approach (this is how Google Maps/Earth or deep-zoom viewers work: they load small tiles at the current zoom level, rather than the entire image at once). During capture, if a system like AWARE takes a gigapixel snapshot, it must stream data from dozens of sensors, potentially at video rates. This requires high-bandwidth electronics and often real-time compression. Brady et al. noted that one of the primary bottlenecks for ubiquitous gigapixel cameras isn't the optics but rather *"lower power and more compact integrated circuits"* to handle all the sensor data (sciencedaily.com). In practical terms, field units like GigaPan store hundreds of JPEG images on memory cards (manageable), but then stitching them yields the big file on a computer. As gigapixel counts rose, researchers have turned to HPC (high-performance computing). The 2018 terapixel macro photo project documented a workflow involving **hundreds of hours of CPU/GPU processing**: sorting images (21 hrs), focus stacking (1,260 hrs), stitching (126 hrs), etc., all before producing the final 1.05 terapixel image (petapixel.com). Clearly, efficient software (parallelized, using GPUs and clusters) is vital. Today's tools like Microsoft ICE or PTGui can stitch a few gigapixels on a high-end desktop, but at the extreme end (hundreds of gigapixels) often custom pipelines or supercomputers are used (petapixel.com). To store and share the output, heavy *compression* is applied – formats like JPEG2000 or advanced wavelet compression can shrink gigapixel files for distribution. There's a trade-off between lossless quality and file size. For online viewing, usually JPEG tiles are used since some quality loss is acceptable for huge gains in size. Engineers have to ensure that despite compression, the fine details (the whole point of gigapixel images) aren't visibly degraded.

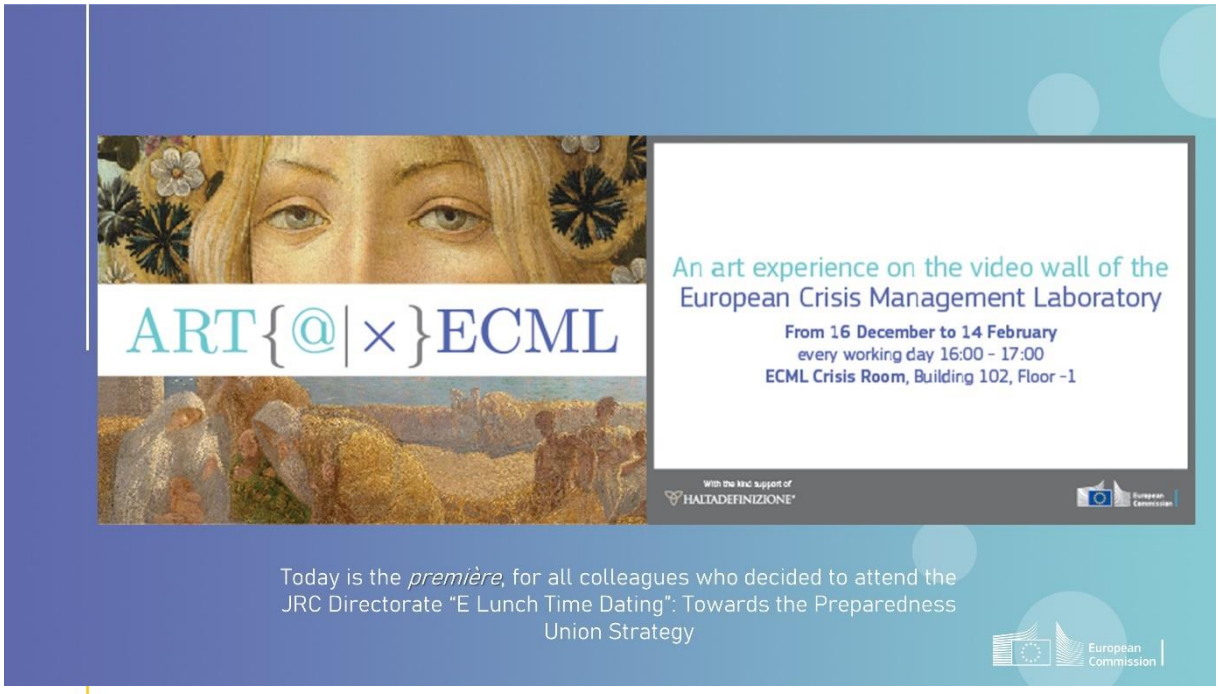
- **Computational Stitching and Software:** At the heart of mosaic-based gigapixel imaging is the stitching software. It must align possibly thousands of images with sub-pixel accuracy. Early on, this was a painstaking manual process, but now algorithms handle it. The software typically performs: feature detection (identifying overlapping points between images), geometric warping (to correct lens distortion and project images into a common projection), blending, and tone matching. One challenge is dealing with *parallax* – if near objects and far objects shift relative to each other between images, you can get stitching errors (ghosting or double images). Properly setting the nodal point on panoramic heads mitigates this for distant scenes. For very large scenes (landscapes), *atmospheric effects* can also vary across frames (heat haze, lighting changes), complicating alignment; some advanced tools can optimize sharpness locally to counter blur. In multi-sensor arrays, the stitching is more about calibration (since each sensor sees a fixed tile of the whole). Here, the computational task might involve deconvolving the slight differences in perspective and ensuring seamless merging at boundaries. The Duke AWARE camera's software, for example, was a custom pipeline that assembled the 98 sub-images in real time and corrected for any vignetting between adjacent microcameras (sciencedaily.com). They leveraged a dedicated FPGA/GPU system to handle the massive data throughput. In microscope gigapixel imaging, software must also deal with scale changes if the stage isn't perfectly level (some focus variation) and often does hierarchical processing (first stitch small groups of tiles, then stitch the groups). The state-of-the-art stitching algorithms (like those based on bundle adjustment from photogrammetry) produce very high-quality panoramas with minimal distortion. Computational photography researchers continue to refine these, for instance using machine learning to better blend seams or detect misalignments. Overall, robust software is as important as hardware in gigapixel imaging – without it, one simply couldn't turn tens of thousands of small images into one coherent whole. The success of projects like the Rijksmuseum's 717 GP *Night Watch* image was in large part due to sophisticated computational processing as much as the capture technique.

- **Physical and Logistical Factors:** There are other pragmatic engineering issues: **stability and motion** – when shooting hundreds of images, even tiny vibrations can ruin some tiles (blur) and throw off the mosaic. GigaPan mounts often must pause for a moment after moving to let vibrations dampen and use mirror lock-up on DSLRs to reduce shake. Long gigs require consistent *power supply* (the GigaPan runs on rechargeable batteries for several hours; bigger custom rigs may need AC power or large battery packs). **Environmental conditions** can be problematic: changing light (moving clouds, setting sun) will cause exposure or colour changes across the panorama – photographers try to shoot gigapixels quickly or under steady lighting (e.g., around noon or on an overcast day) to avoid a patchwork look. In some cases, a panorama might even span day to night, which advanced editing can correct but is difficult. For aerial or satellite gigapixel imaging, motion and vibration are even more extreme (stabilized platforms and fast exposure times are needed). **Thermal management** is a concern for multi-sensor cameras: packing dozens of sensors and boards in a box leads to heat – AWARE cameras consumed

hundreds of watts of power and had fans to keep components cool (distantfocus.com), since hot sensors produce more noise.

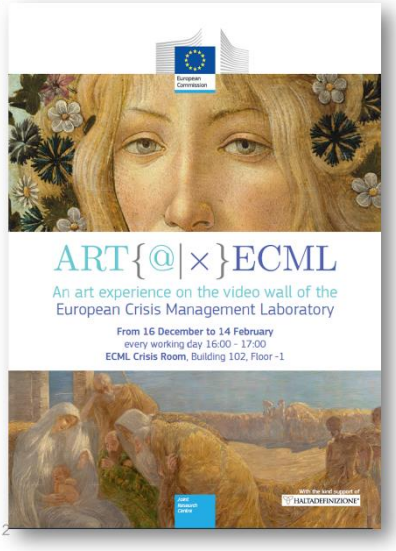
- **Viewing and Distribution:** Once you have a gigapixel image, how do you share it? This is more of a software challenge than hardware, but worth noting. Specialized viewers (Zoomify, OpenSeadragon, krpano, etc.) were developed to allow smooth zoom and pan of giant images via the internet. These work by creating multi-resolution tile sets of the image. For interactive exhibits, sometimes custom GPU-based viewers are used on high-res displays or VR. The *user experience* of gigapixel images has driven some technical adjustments – for instance, adding annotation or search capability (finding a person in a crowd GigaPan, etc.) required metadata handling at massive scale. Companies have offered cloud services to host gigapixel images (because hosting a 20 GB image is non-trivial for average users). Thus, an ecosystem of software tools (from capture to post to viewing) had to mature alongside the cameras.

Annex 8 – Exhibition presentation



Art {@|×} ECML

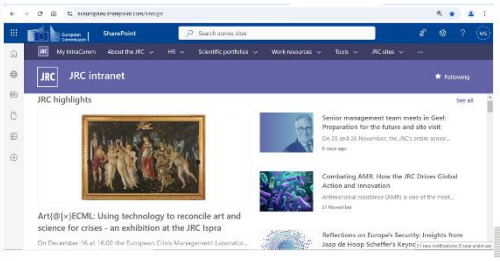
Technology to reconcile art and science



An innovative digital technology, specifically designed to manage digital art-historical heritage in very high definition.

An experiment to explore its adaptability to DRM and other JRC fields of research.

An opportunity to enjoy an immersive experience and discover world-famous art masterpieces



2

European Commission

Reimagining crises through the lens of art and technology

This collaboration draws inspiration from **Camerino, a medieval Italian town devastated by earthquakes in 1997 and 2016**, which used Haltadefinizione's technology to digitalize and enhance its cultural heritage.



The Camerino exhibition's success in **merging art and technology to address the emotional and cultural toll of crises** inspired the idea of Art{@|×}ECML.

The vision is to replicate this synergy at the **ECML, temporarily transforming it into a space where science, technology and art can inform and elevate one another.**

European Commission

Zoom into fine details that are almost invisible to the naked eye

Cultural heritage preservation

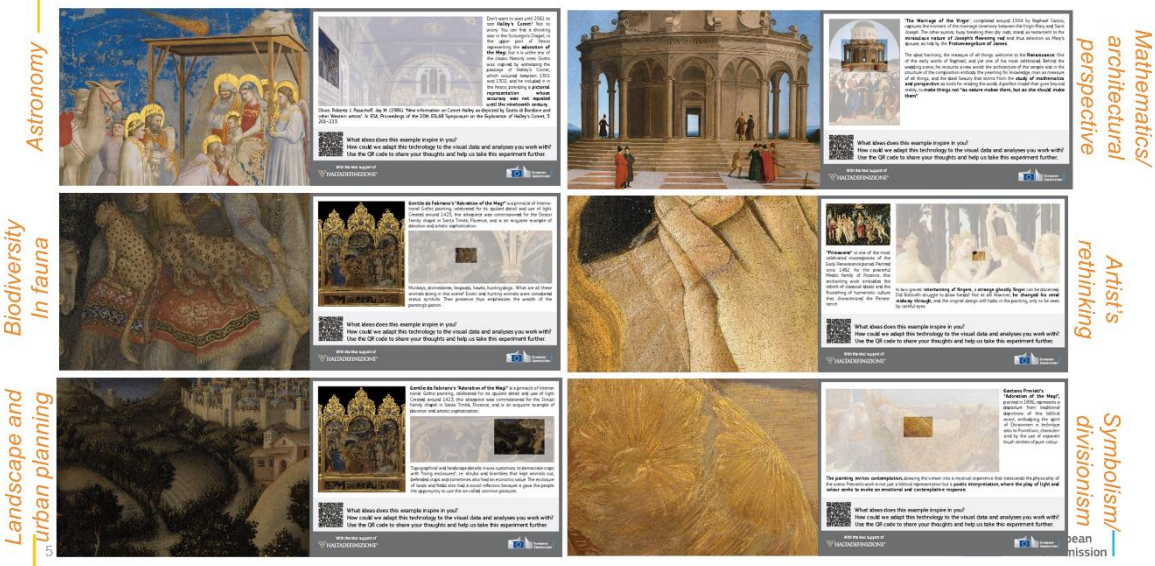




Biodiversity/flora

European Commission

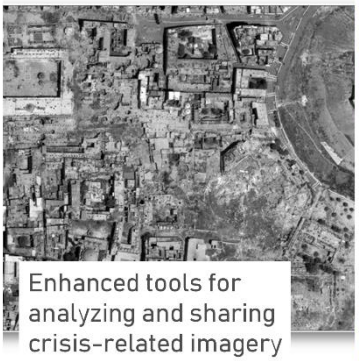
Art, science and technology were naturally intertwined in the past



5

Why this project matters

The exhibition **explores synergies between art, technology and crisis management**.
By applying this or a similar technology to the ECML's tools, this pilot could pave the way for:



Enhanced tools for analyzing and sharing crisis-related imagery



Studying crisis perception through collaborations with behavioral experts



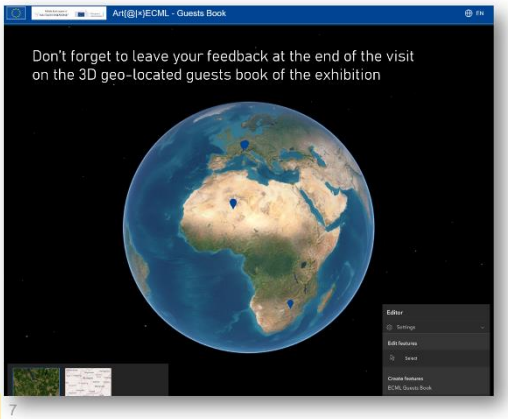
Preserving cultural heritage, documenting how crises impact art.

6

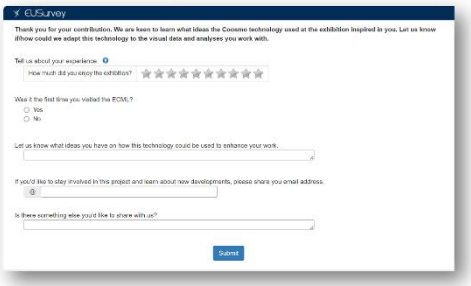
European Commission

A 2 months-long JRC-wide brainstorming

What **ideas** does this example inspire in you? How could we **adapt this technology** to the visual **data** and analyses you work with? *Share your thoughts and help us take this experiment further!*



Link to EU Survey: https://ec.europa.eu/eusurvey/runner/d373cae3-f0b3-9cba-36a0-44402fc6f840



7

European Commission

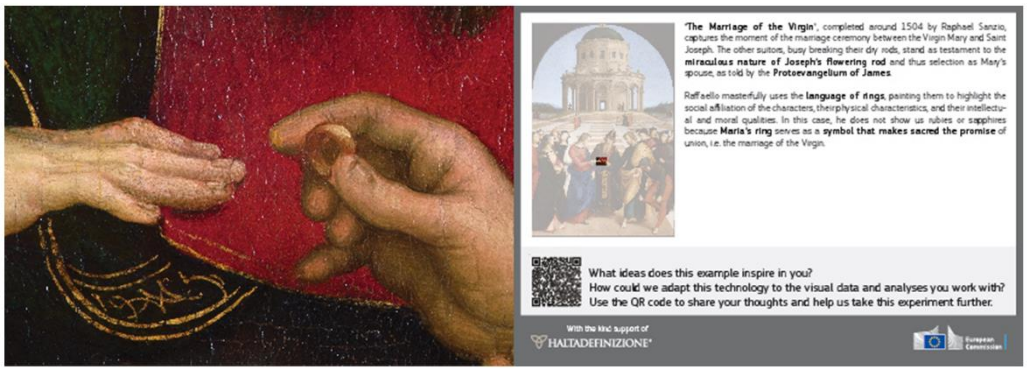
To conclude, this initiative will allow us to…

- **test** innovative **digital technologies** in the ECML
- **increase** the **usability** of the video wall (a similar one is being installed in the ERCC)
- **collect ideas at JRC-wide level** for innovative use of digital technologies
- **inspire** how to **communicate** differently our science and knowledge
- **attract** colleagues to the ECML, thus increasing its **visibility**
- **experience** the collaboration with a digital tech company, thus experiencing how it could be if we work to become a **JRC open lab**
- **enjoy** an immersive experience of masterpieces from the past, thus hopefully contributing to our **well being** ☺

8

European Commission

Art & SciTech: a marriage made in heaven



European Commission

Thank you for your attention

Big thanks to all contributors for the good work done in record time: Astarita Simone, Bortolamei Fabio, De Girolamo Ludovica, Galliano Daniele Alberto, Mancon Lorenzo, Moreira Agrela Andre, Sabbatino Emanuele, Spagnolo Luigi and the whole ECML team for the support, feedbacks and inspirational ideas; Haltadefinizione, a company of the Group Panini Cultura.

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European Commission

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